

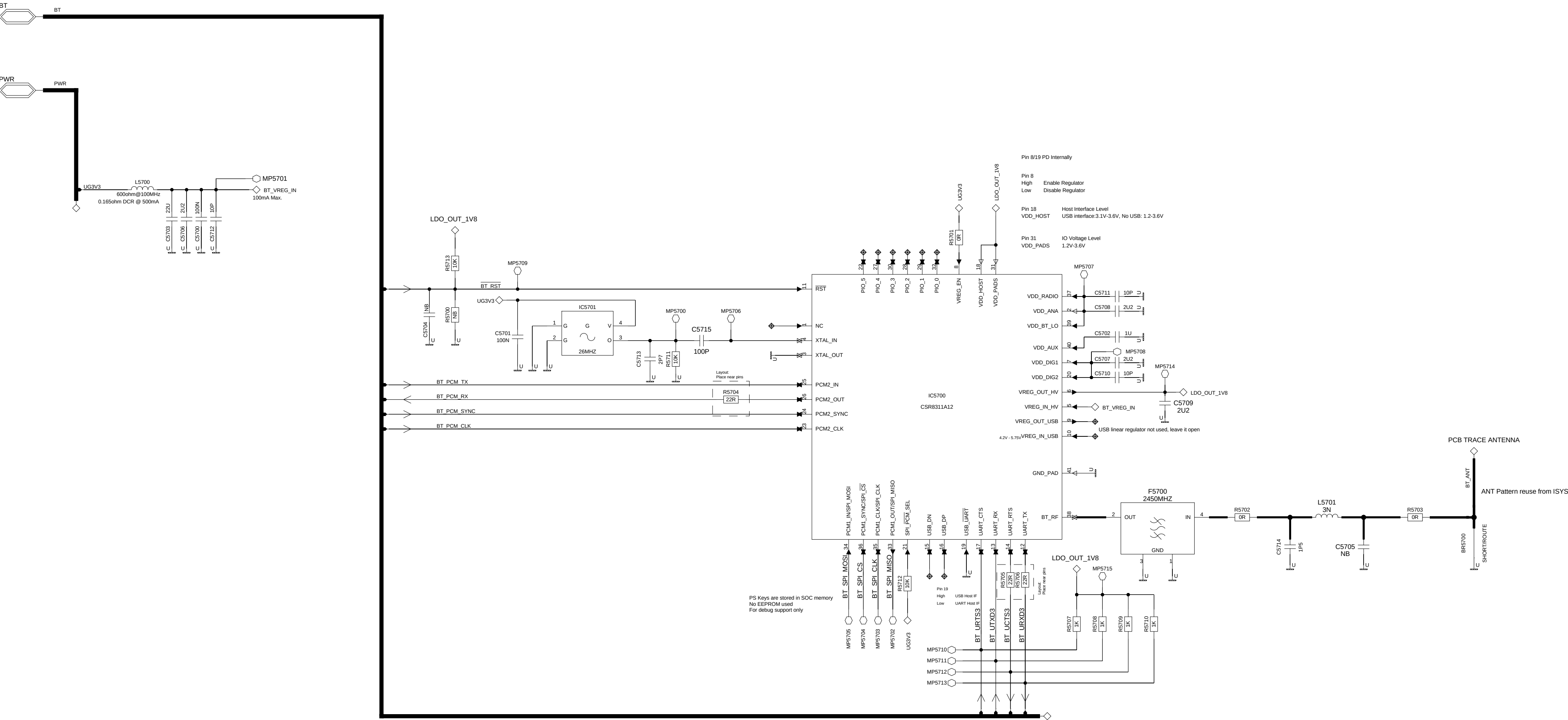
BARCODE: LABEL65X65
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BARCODE: LABEL65X65

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A	CN00687 CN006912	19-12-23 07-30-2023	R	GZ
Rev	ED 11mkt release			

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	created	12-20-2023		
	checked	12-20-2023		

Sample	Sheets of	Sheet
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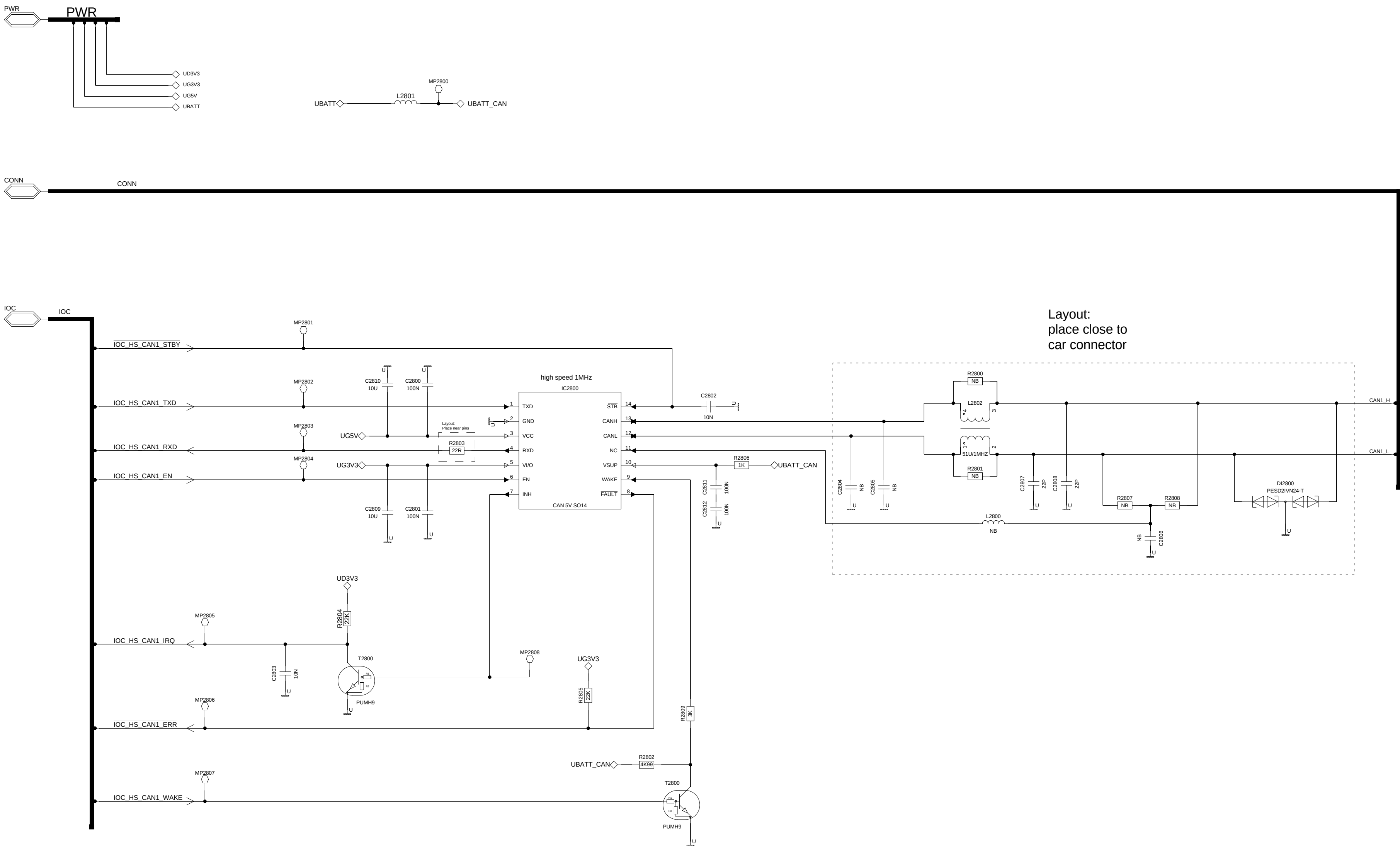
BLUETOOTH (5700-5799)



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Rev.	type of change		change-no.	date of change	auth. checked
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4010_Bluetooth			2	31	
substitute for:			Type	Variant	Part Number
				C143	

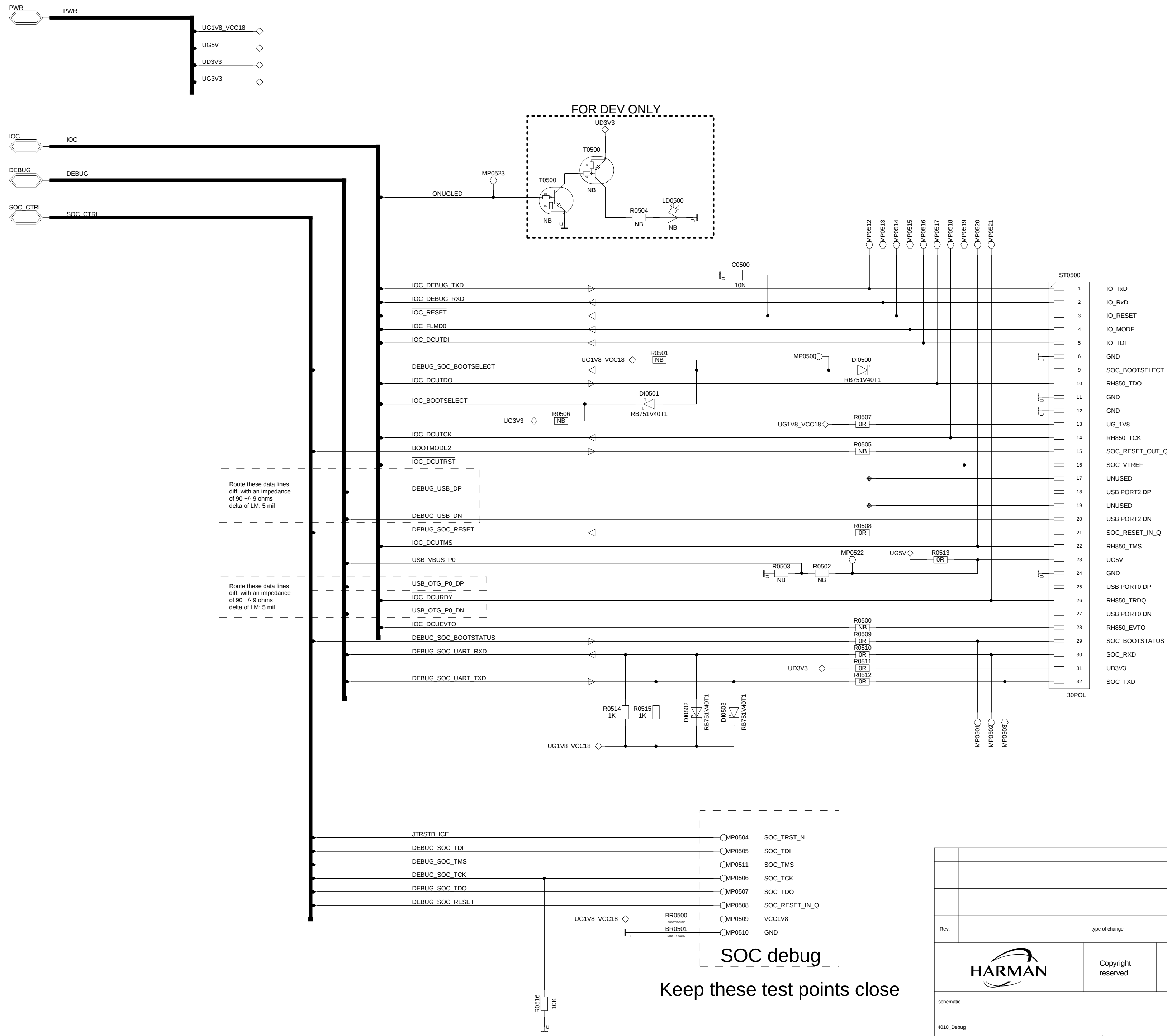
CAN (2800-2899)



Layout:
place close to
car connector

Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet	Sheets of	Sample
4010_CAN			3	31	
substitute for:		substitute by:	Type	Variant	Part Number

The schematic diagram illustrates the ATMX1057T-A19 board, centered around the IC7000. The board includes a power supply section with a 3.3V regulator (MP7000) and a 1.8V regulator (MP7005). The IC7000 is connected to various pins, including TP_RST, TP_RSTQ, TP_SCL, TP_SDA, TP_SYNC, and TP_SYNCQ. The board also features a USB-to-UART bridge (ATMX1057T-A19) and a USB-to-UART bridge (ATMX1057T-A19). The board is populated with various components, including resistors (R7000, R7001, R7002, R7003, R7004, R7005, R7006, R7007, R7008, R7009, R7010, R7011, R7012, R7013, R7014, R7015, R7016, R7017, R7018, R7019, R7020, R7021, R7022, R7023, R7024, R7025, R7026, R7027, R7028, R7029, R7030, R7031, R7032, R7033, R7034, R7035, R7036, R7037, R7038, R7039, R7040, R7041, R7042, R7043, R7044, R7045, R7046, R7047, R7048, R7049, R7050, R7051, R7052, R7053, R7054, R7055, R7056, R7057, R7058, R7059, R7060, R7061, R7062, R7063, R7064, R7065, R7066, R7067, R7068, R7069, R7070, R7071, R7072, R7073, R7074, R7075, R7076, R7077, R7078, R7079, R7080, R7081, R7082, R7083, R7084, R7085, R7086, R7087, R7088, R7089, R7090, R7091, R7092, R7093, R7094, R7095, R7096, R7097, R7098, R7099, R7100, R7101, R7102, R7103, R7104, R7105, R7106, R7107, R7108, R7109, R7110, R7111, R7112, R7113, R7114, R7115, R7116, R7117, R7118, R7119, R7120, R7121, R7122, R7123, R7124, R7125, R7126, R7127, R7128, R7129, R7130, R7131, R7132, R7133, R7134, R7135, R7136, R7137, R7138, R7139, R7140, R7141, R7142, R7143, R7144, R7145, R7146, R7147, R7148, R7149, R7150, R7151, R7152, R7153, R7154, R7155, R7156, R7157, R7158, R7159, R7160, R7161, R7162, R7163, R7164, R7165, R7166, R7167, R7168, R7169, R7170, R7171, R7172, R7173, R7174, R7175, R7176, R7177, R7178, R7179, R7180, R7181, R7182, R7183, R7184, R7185, R7186, R7187, R7188, R7189, R7190, R7191, R7192, R7193, R7194, R7195, R7196, R7197, R7198, R7199, R7200, R7201, R7202, R7203, R7204, R7205, R7206, R7207, R7208, R7209, R7210, R7211, R7212, R7213, R7214, R7215, R7216, R7217, R7218, R7219, R7220, R7221, R7222, R7223, R7224, R7225, R7226, R7227, R7228, R7229, R7230, R7231, R7232, R7233, R7234, R7235, R7236, R7237, R7238, R7239, R7240, R7241, R7242, R7243, R7244, R7245, R7246, R7247, R7248, R7249, R7250, R7251, R7252, R7253, R7254, R7255, R7256, R7257, R7258, R7259, R7260, R7261, R7262, R7263, R7264, R7265, R7266, R7267, R7268, R7269, R7270, R7271, R7272, R7273, R7274, R7275, R7276, R7277, R7278, R7279, R7280, R7281, R7282, R7283, R7284, R7285, R7286, R7287, R7288, R7289, R7290, R7291, R7292, R7293, R7294, R7295, R7296, R7297, R7298, R7299, R7300, R7301, R7302, R7303, R7304, R7305, R7306, R7307, R7308, R7309, R7310, R7311, R7312, R7313, R7314, R7315, R7316, R7317, R7318, R7319, R7320, R7321, R7322, R7323, R7324, R7325, R7326, R7327, R7328, R7329, R7330, R7331, R7332, R7333, R7334, R7335, R7336, R7337, R7338, R7339, R7340, R7341, R7342, R7343, R7344, R7345, R7346, R7347, R7348, R7349, R7350, R7351, R7352, R7353, R7354, R7355, R7356, R7357, R7358, R7359, R7360, R7361, R7362, R7363, R7364, R7365, R7366, R7367, R7368, R7369, R7370, R7371, R7372, R7373, R7374, R7375, R7376, R7377, R7378, R7379, R7380, R7381, R7382, R7383, R7384, R7385, R7386, R7387, R7388, R7389, R7390, R7391, R7392, R7393, R7394, R7395, R7396, R7397, R7398, R7399, R7400, R7401, R7402, R7403, R7404, R7405, R7406, R7407, R7408, R7409, R7410, R7411, R7412, R7413, R7414, R7415, R7416, R7417, R7418, R7419, R7420, R7421, R7422, R7423, R7424, R7425, R7426, R7427, R7428, R7429, R7430, R7431, R7432, R7433, R7434, R7435, R7436, R7437, R7438, R7439, R7440, R7441, R7442, R7443, R7444, R7445, R7446, R7447, R7448, R7449, R7450, R7451, R7452, R7453, R7454, R7455, R7456, R7457, R7458, R7459, R7460, R7461, R7462, R7463, R7464, R7465, R7466, R7467, R7468, R7469, R7470, R7471, R7472, R7473, R7474, R7475, R7476, R7477, R7478, R7479, R7480, R7481, R7482, R7483, R7484, R7485, R7486, R7487, R7488, R7489, R7490, R7491, R7492, R7493, R7494, R7495, R7496, R7497, R7498, R7499, R7500, R7501, R7502, R7503, R7504, R7505, R7506, R7507, R7508, R7509, R7510, R7511, R7512, R7513, R7514, R7515, R7516, R7517, R7518, R7519, R7520, R7521, R7522, R7523, R7524, R7525, R7526, R7527, R7528, R7529, R7530, R7531, R7532, R7533, R7534, R7535, R7536, R7537, R7538, R7539, R7540, R7541, R7542, R7543, R7544, R7545, R7546, R7547, R7548, R7549, R7550, R7551, R7552, R7553, R7554, R7555, R7556, R7557, R7558, R7559, R7560, R7561, R7562, R7563, R7564, R7565, R7566, R7567, R7568, R7569, R7570, R7571, R7572, R7573, R7574, R7575, R7576, R7577, R7578, R7579, R7580, R7581, R7582, R7583, R7584, R7585, R7586, R7587, R7588, R7589, R7590, R7591, R7592, R7593, R7594, R7595, R7596, R7597, R7598, R7599, R7600, R7601, R7602, R7603, R7604, R7605, R7606, R7607, R7608, R7609, R7610, R7611, R7612, R7613, R7614, R7615, R7616, R7617, R7618, R7619, R7620, R7621, R7622, R7623, R7624, R7625, R7626, R7627, R7628, R7629, R7630, R7631, R7632, R7633, R7634, R7635, R7636, R7637, R7638, R7639, R7640, R7641, R7642, R7643, R7644, R7645, R7646, R7647, R7648, R7649, R7650, R



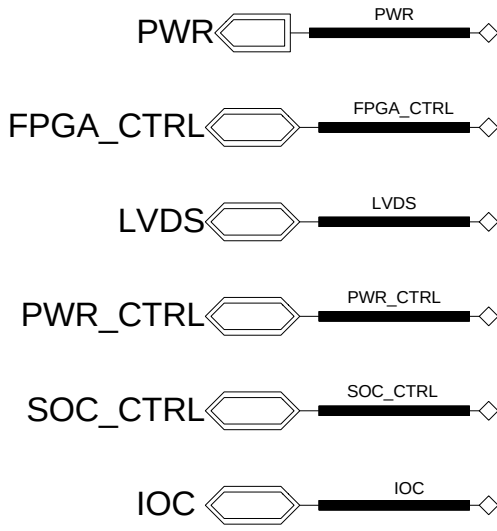
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schematic		Sheet	Sheets of	Sample	
4010_Debug		5	31		
substitute for:		Type	Variant	Part Number	
		substitute by:			

Marvell recommends using Varistors
for ESD protection.
TDK part # AVRH10C101KT1R1NE8

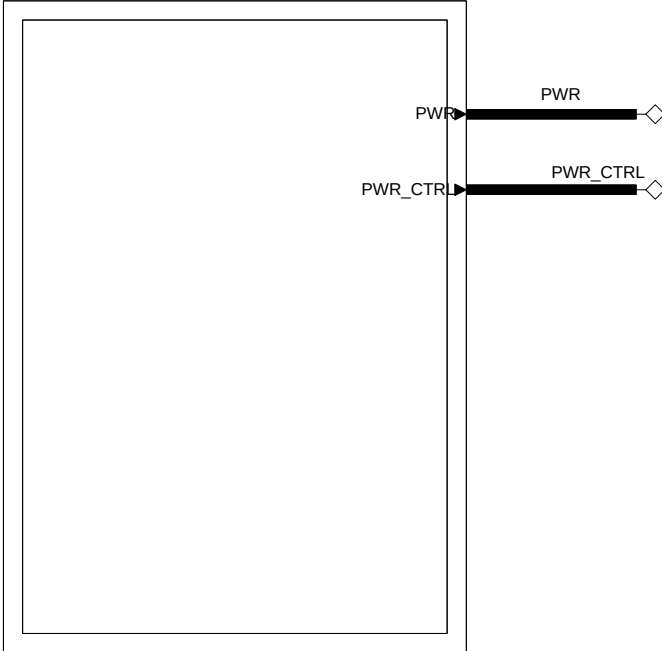
The diagrams show the power supply and ground connections for the UG3V3, UG3V3_ETH_1010, MP8301, MP8302, and MP8303 components. Each diagram includes a 3.3V supply, a 22uF capacitor, and a 1uF capacitor connected to ground.

- UG3V3:** A 3.3V supply connected to a 22uF capacitor (C8312) and a 1uF capacitor (C8305) connected to ground.
- UG3V3_ETH_1010:** A 3.3V supply connected to a 22uF capacitor (C8312) and a 1uF capacitor (C8306) connected to ground.
- MP8301:** A 3.3V supply connected to a 22uF capacitor (C8312) and a 1uF capacitor (C8309) connected to ground.
- MP8302:** A 3.3V supply connected to a 22uF capacitor (C8310) and a 1uF capacitor (C8303) connected to ground.
- MP8303:** A 3.3V supply connected to a 22uF capacitor (C8311) and a 1uF capacitor (C8304) connected to ground.

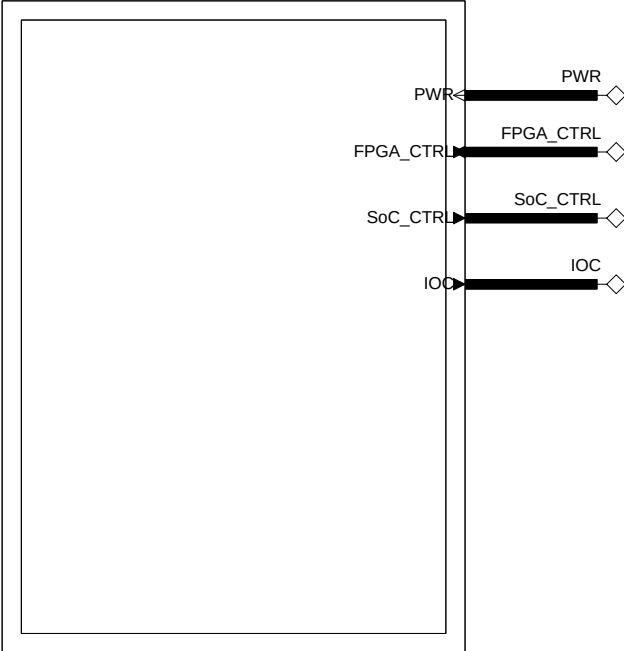
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4010_ETH_PHY			6	31	
substitute for:		substitute by:	Type	Variant	Part Number



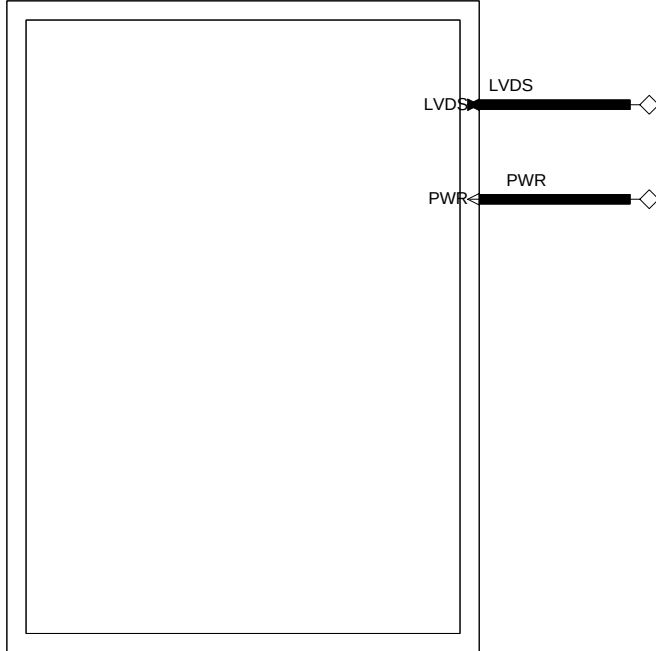
FPGA_POWER



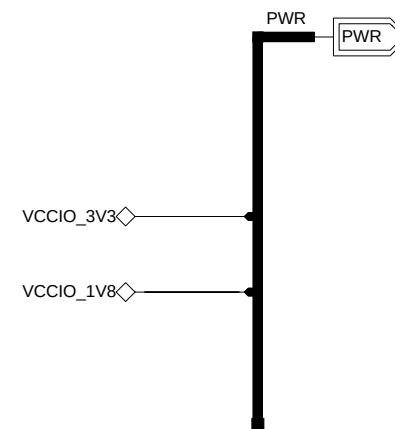
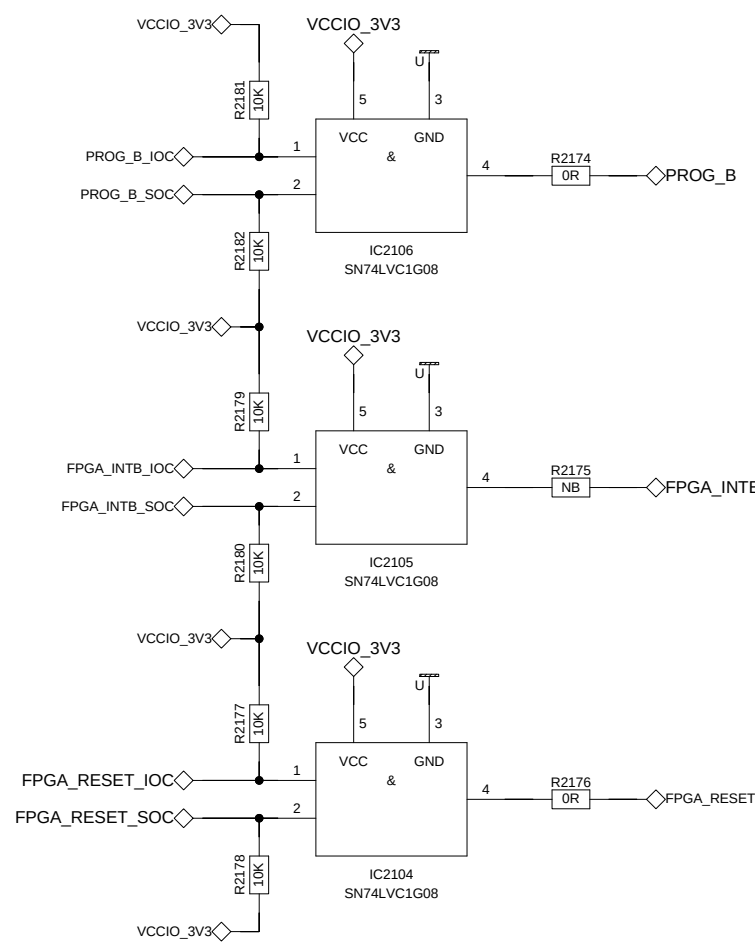
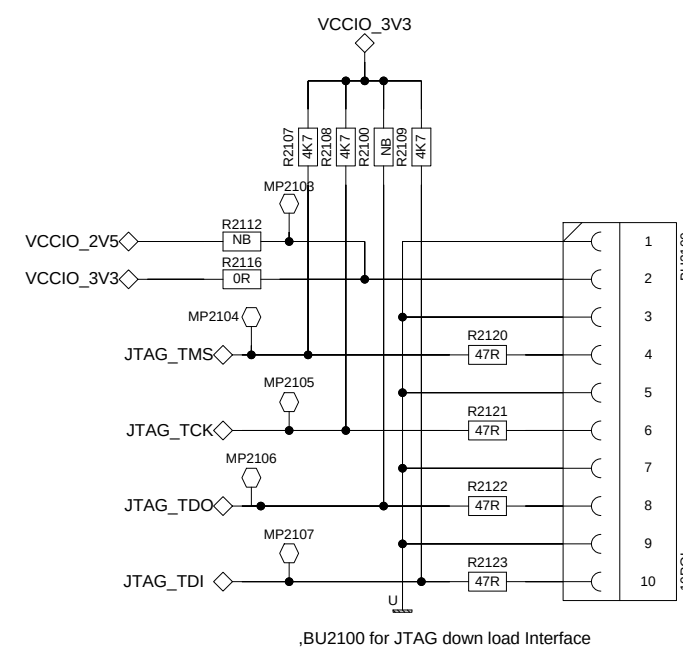
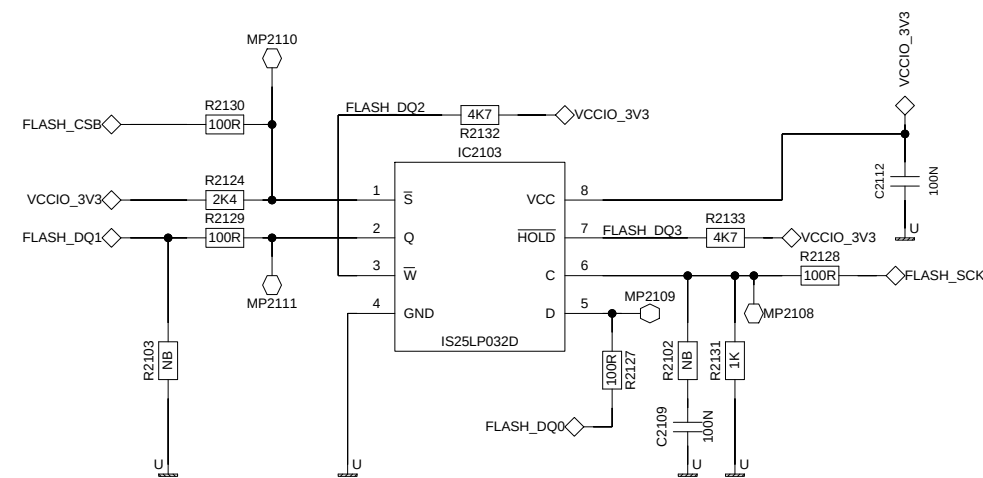
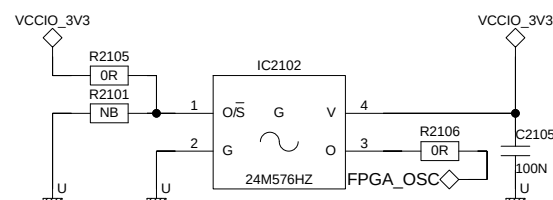
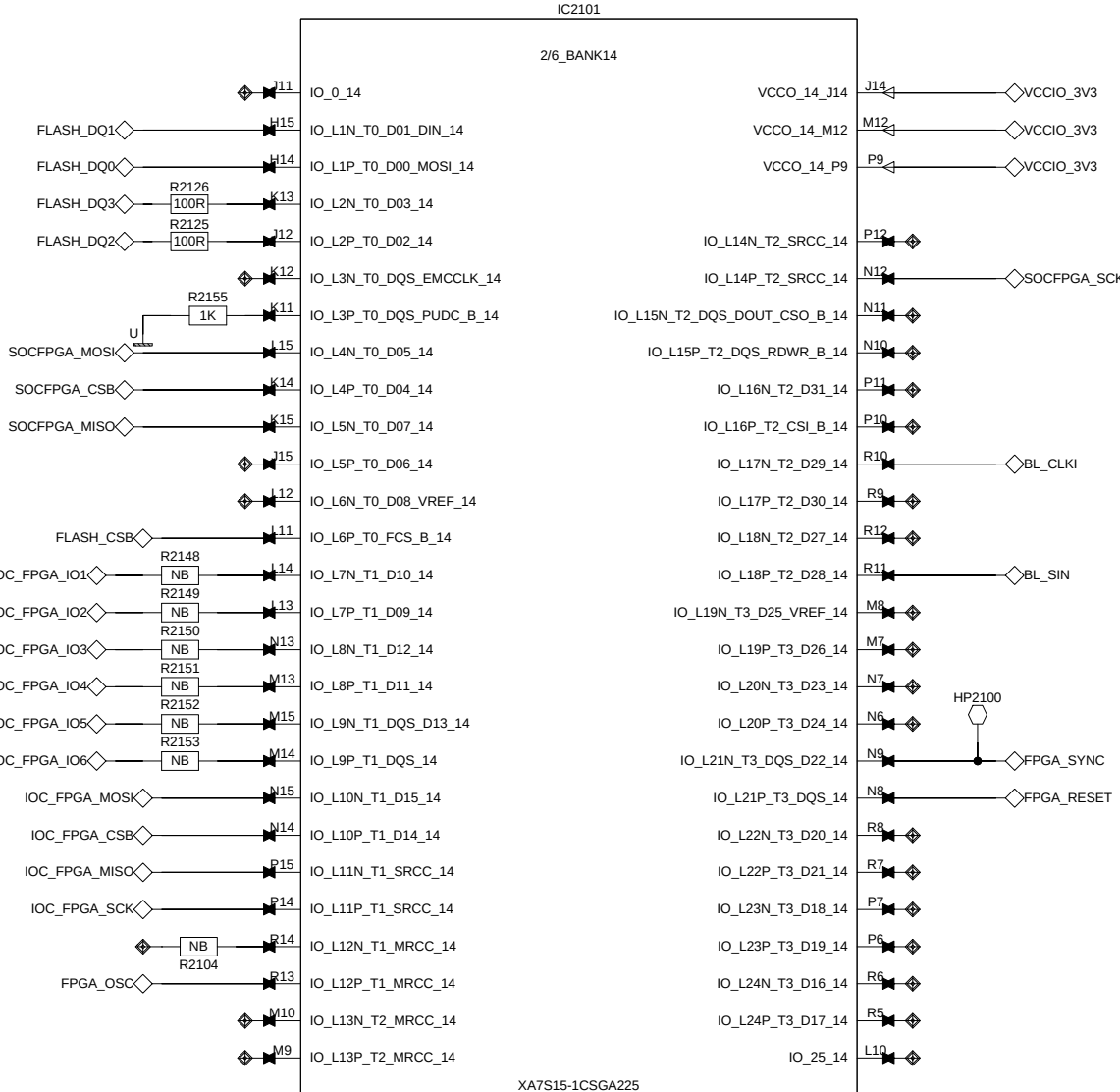
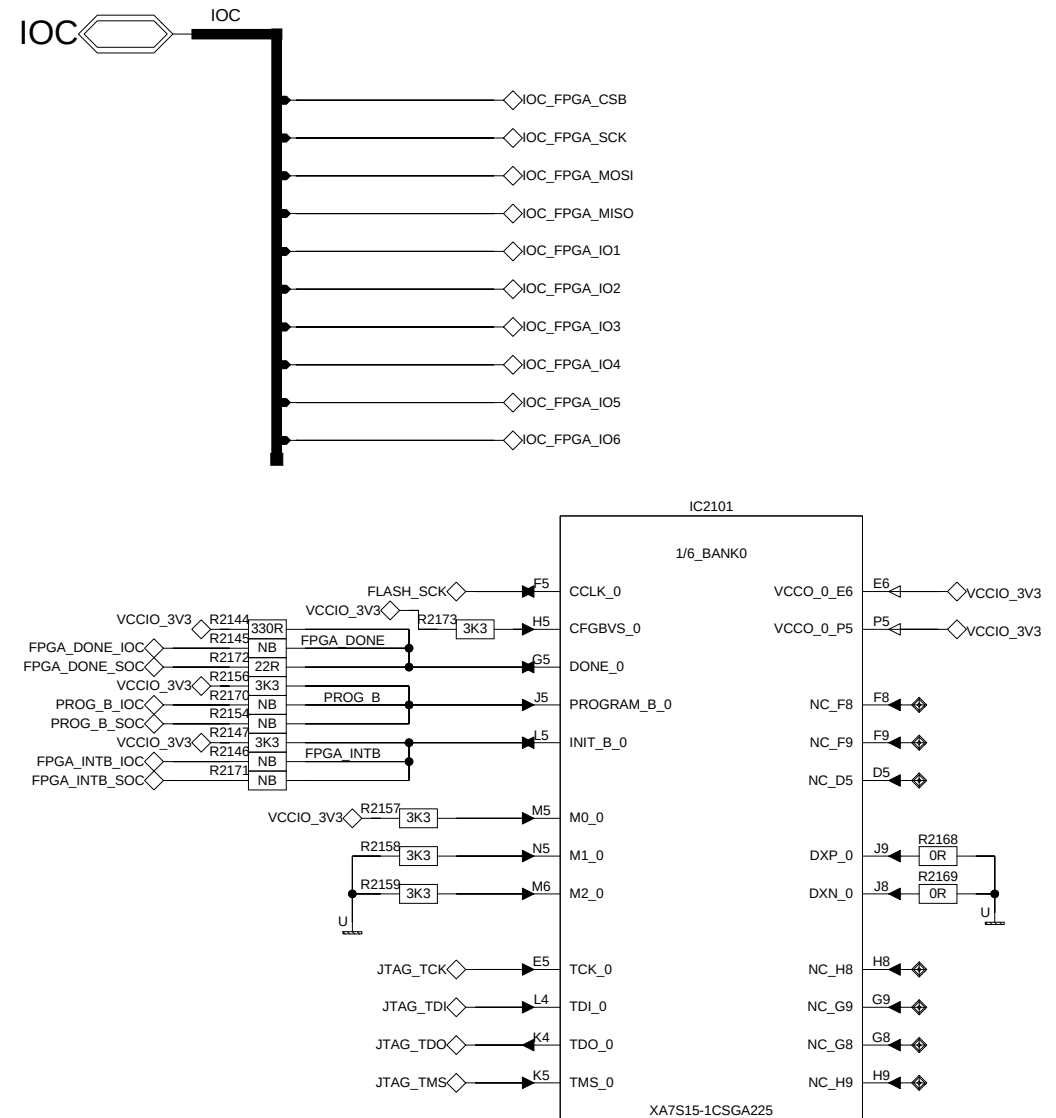
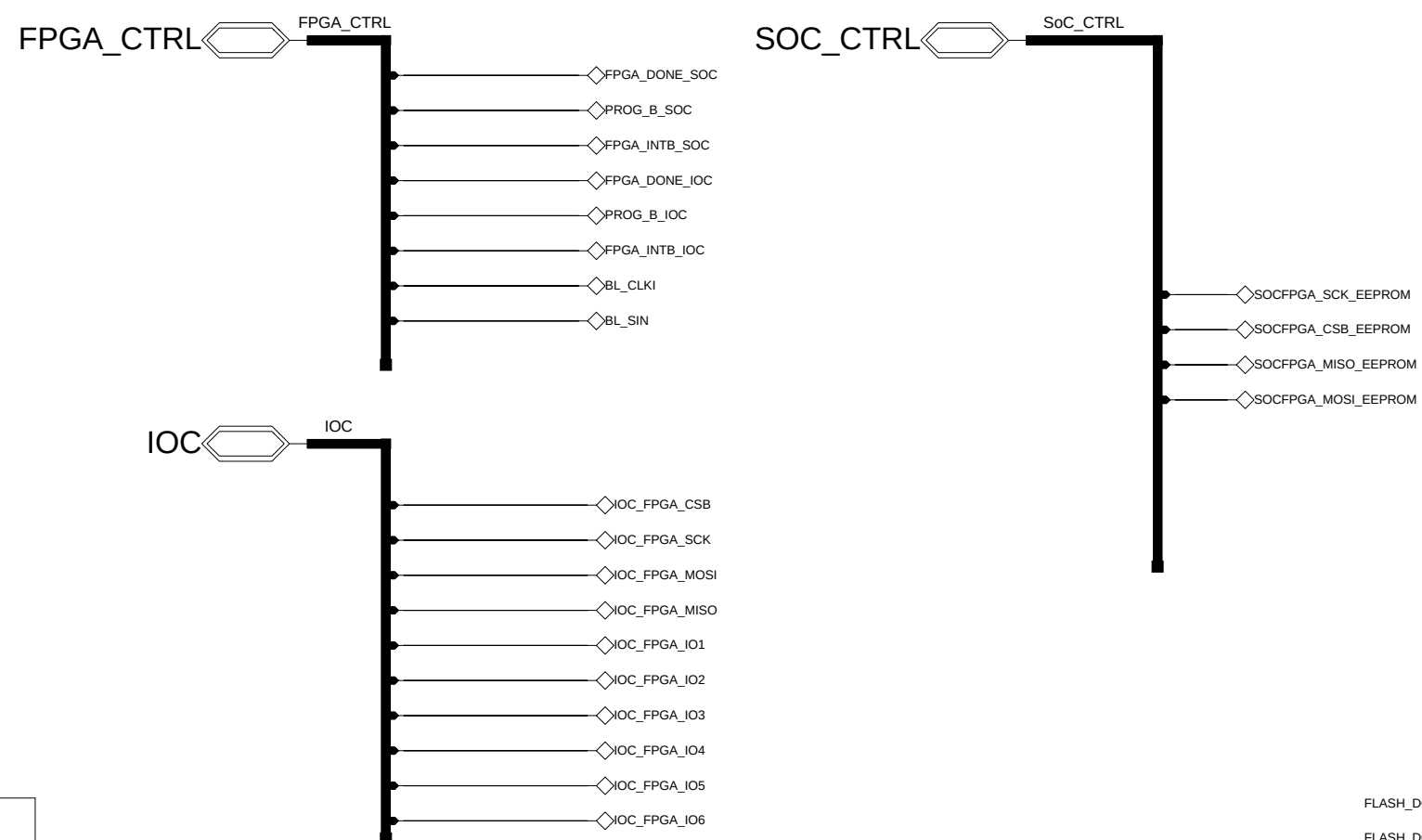
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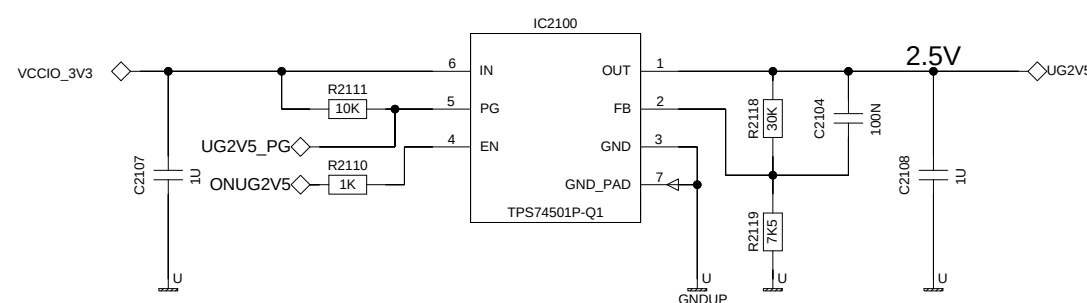
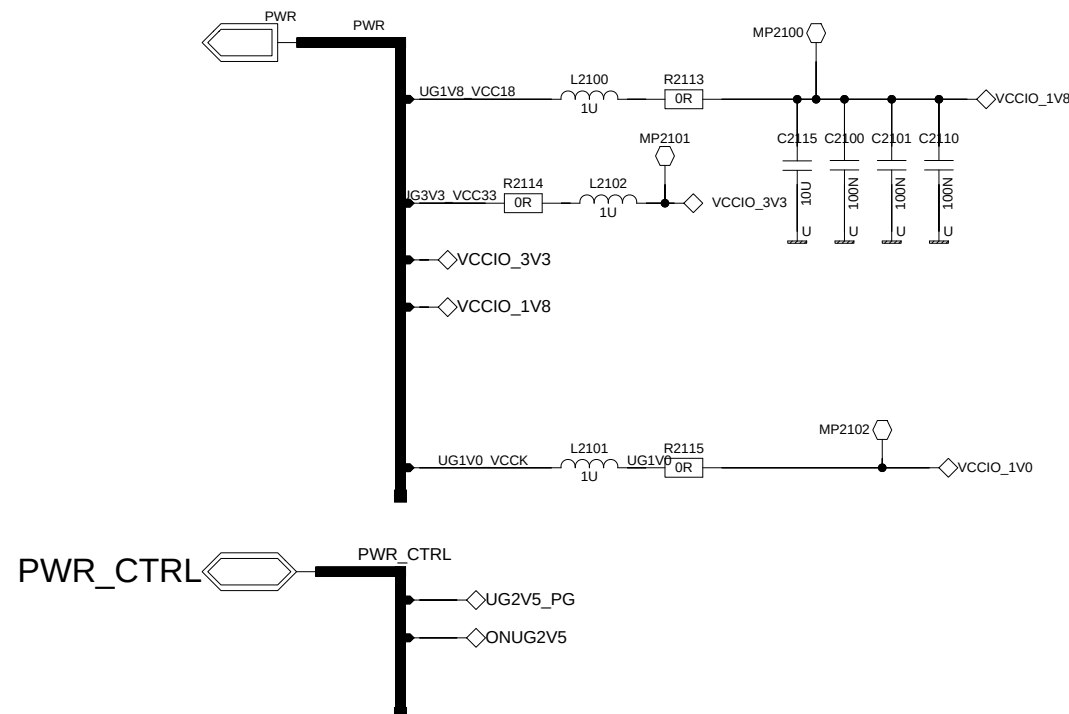
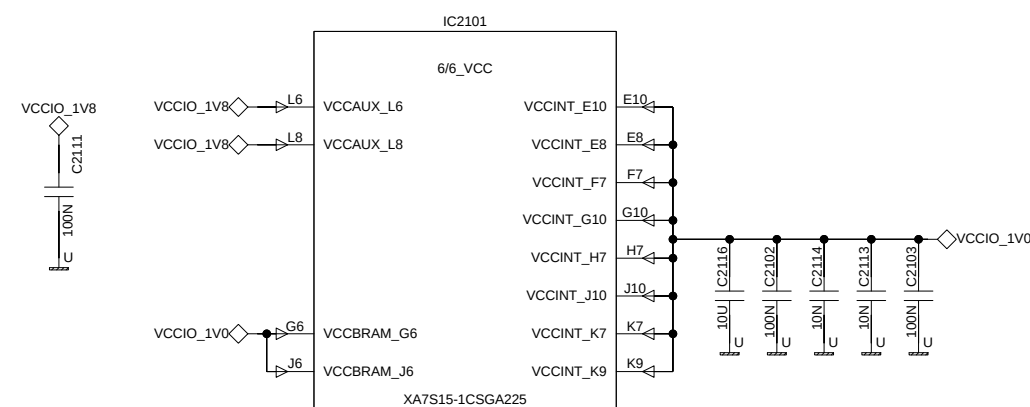
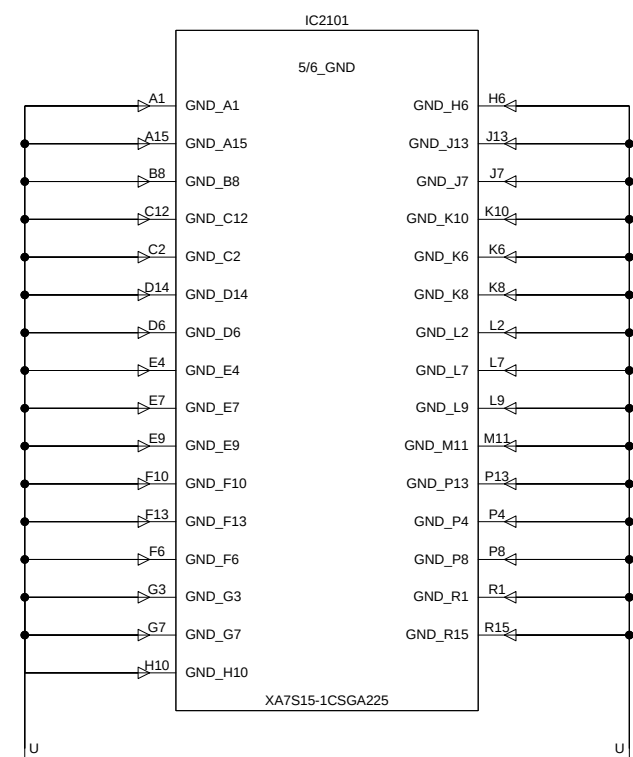
FPGA_VIDEO



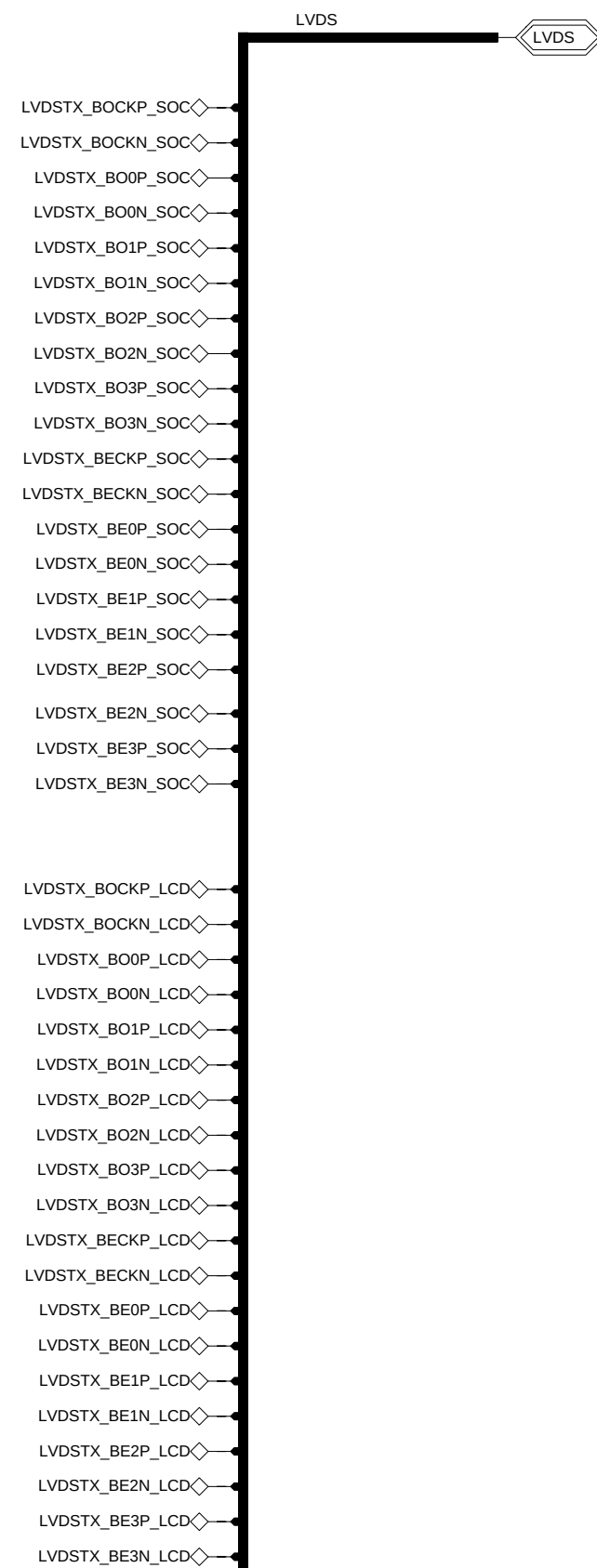
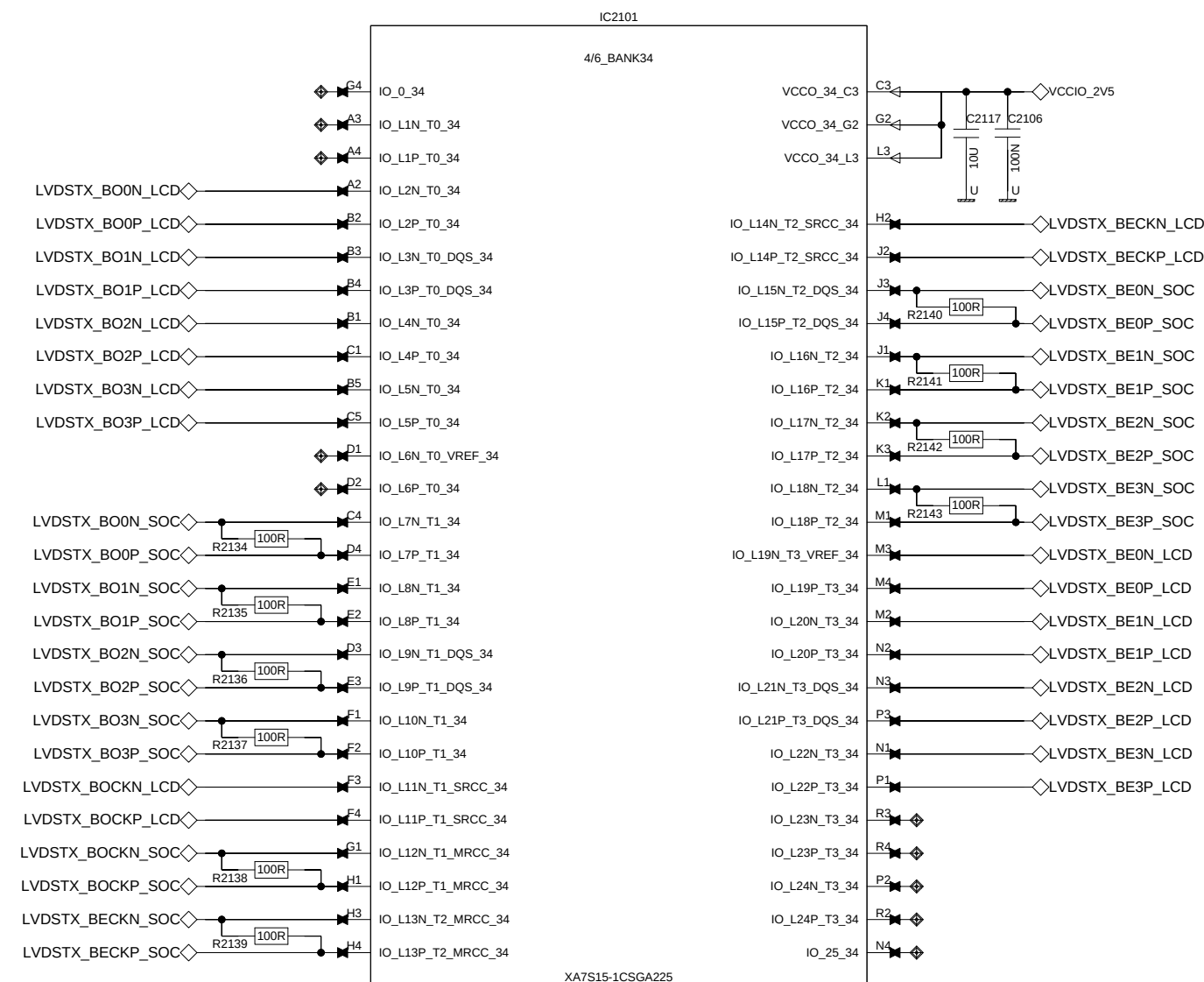
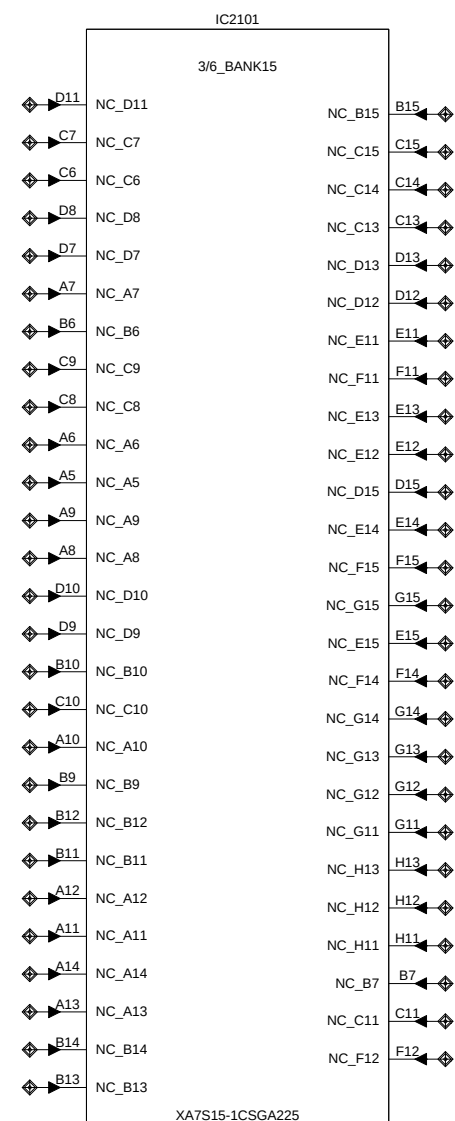
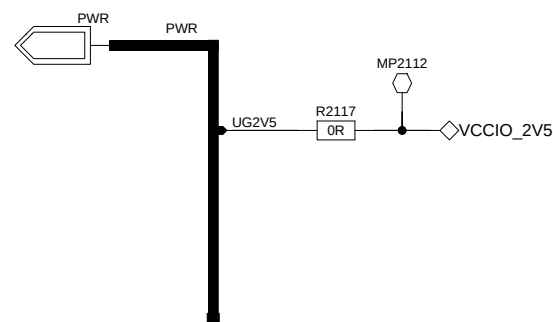
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4030_FPGA			Type	Variant C143	Part Number
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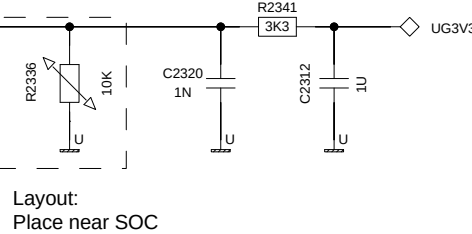
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FPGA_CONF			Type	Variant	Part Number
substitute for:		substitute by:		C143	



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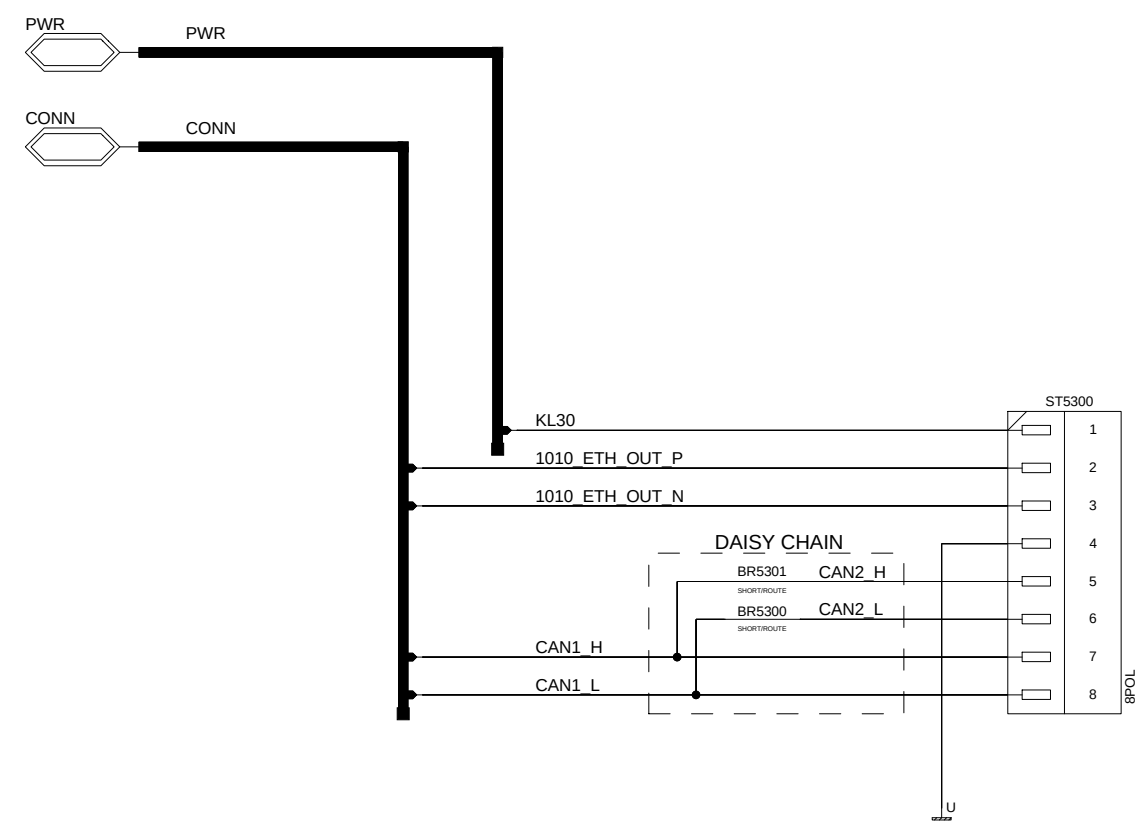
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schematic			Sheet	Sheets of	Sample
FPGA_VIDEO			10	31	
substitute for:			Type	Variant	Part Number
				C143	



RH550 INTERRUPT TABLE		
INTP0	AWO	IOC_BKLT_FAULT
INTP1	ISO	
INTP2	ISO	
INTP3	ISO	HS_CAN1_RXD
INTP4	ISO	-----
INTP5	AWO	-----
INTP6	AWO	-----
INTP7	AWO	
INTP8	AWO	HS_CAN1_IRQ
INTP9	AWO	
INTP10	ISO	IOC_SOC_IPC_IRQ
INTP11	AWO	-----
INTP12	AWO	-----
INTP13	AWO	
INTP14	AWO	DEBUG_RX
INTP15	AWO	

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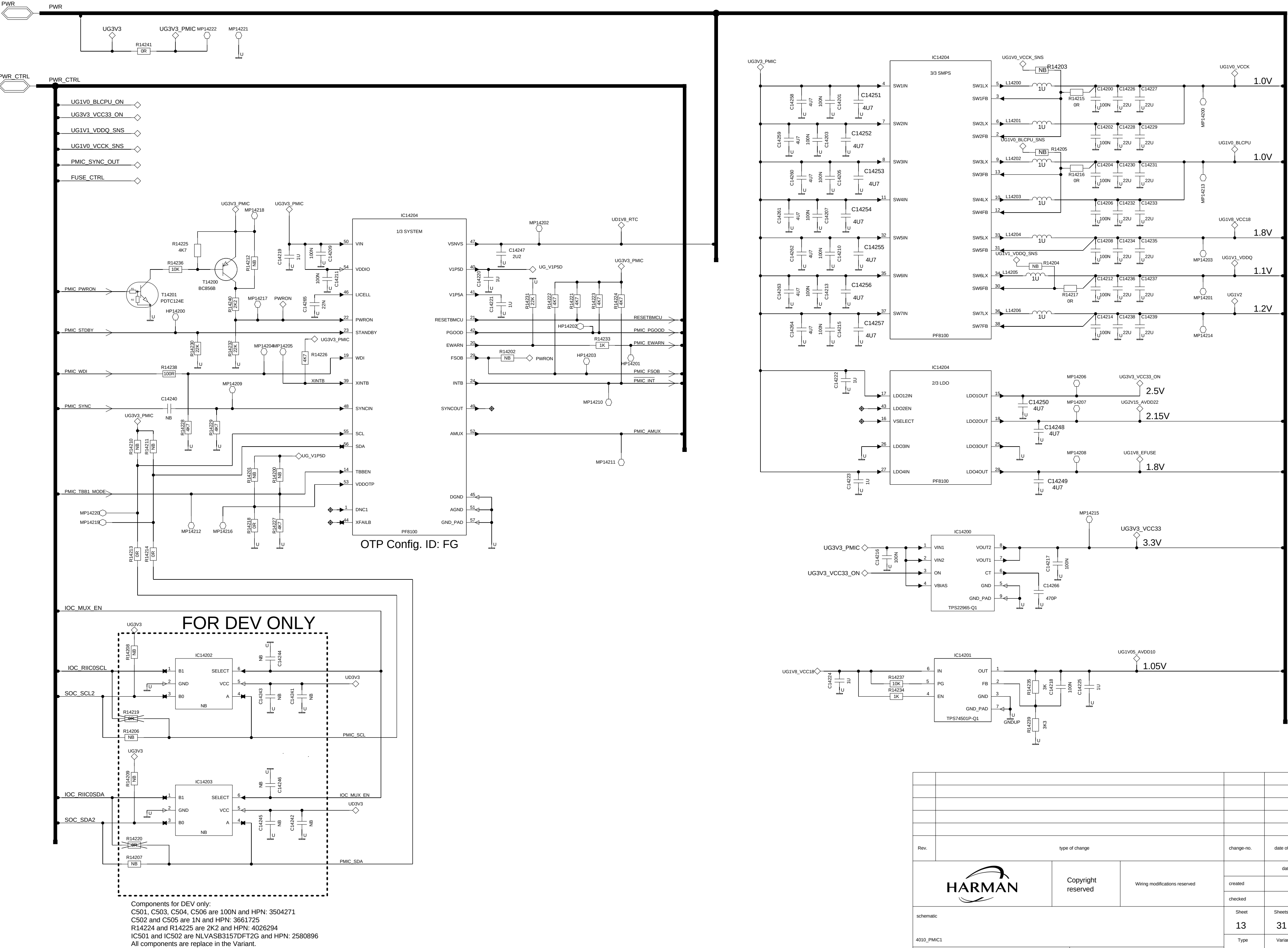
MAIN CONNECTOR (5300-5399)



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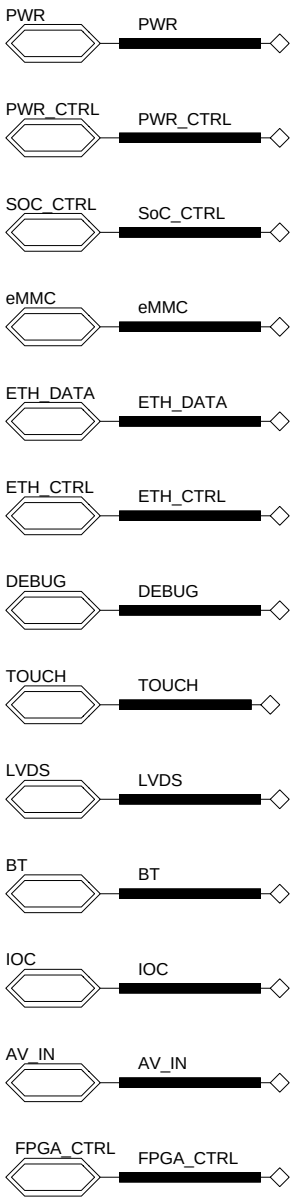
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substitute by:						

PMIC1 (14200-14299)

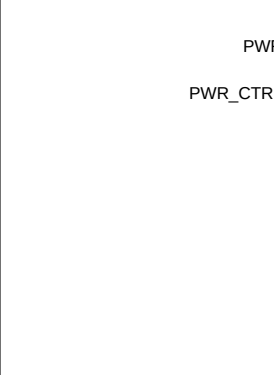


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SOC (10000-10999)

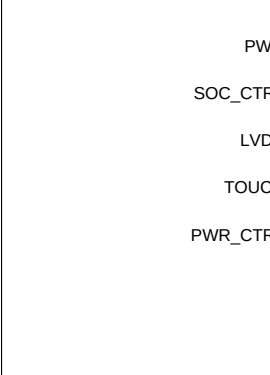


4010 SoC Power



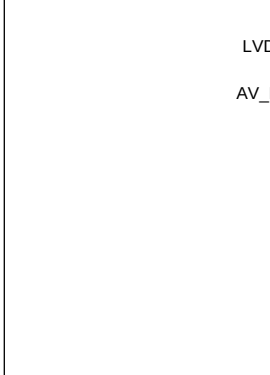
10500 - 10599

4010 SoC_CTRL



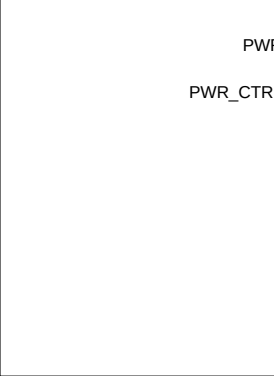
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4010 SoC Video



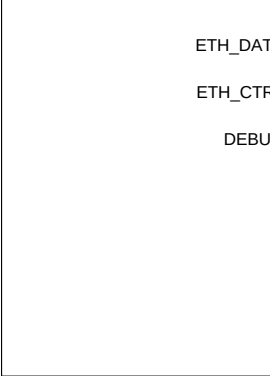
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4010 SoC LPDDR4



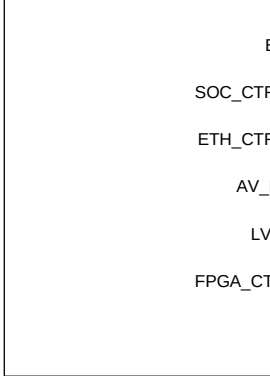
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4010 SoC_ETH_PCIE_USB



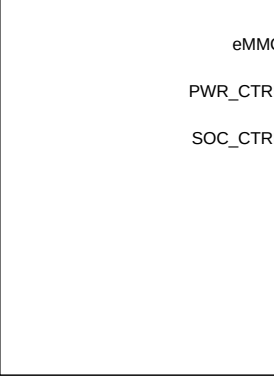
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4010 SoC Audio



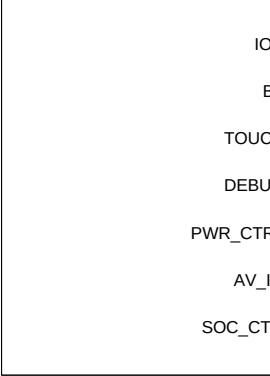
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4010 SoC_eMMC_NAND_SD



10100 - 10199

4010 SoC_Serial_IF

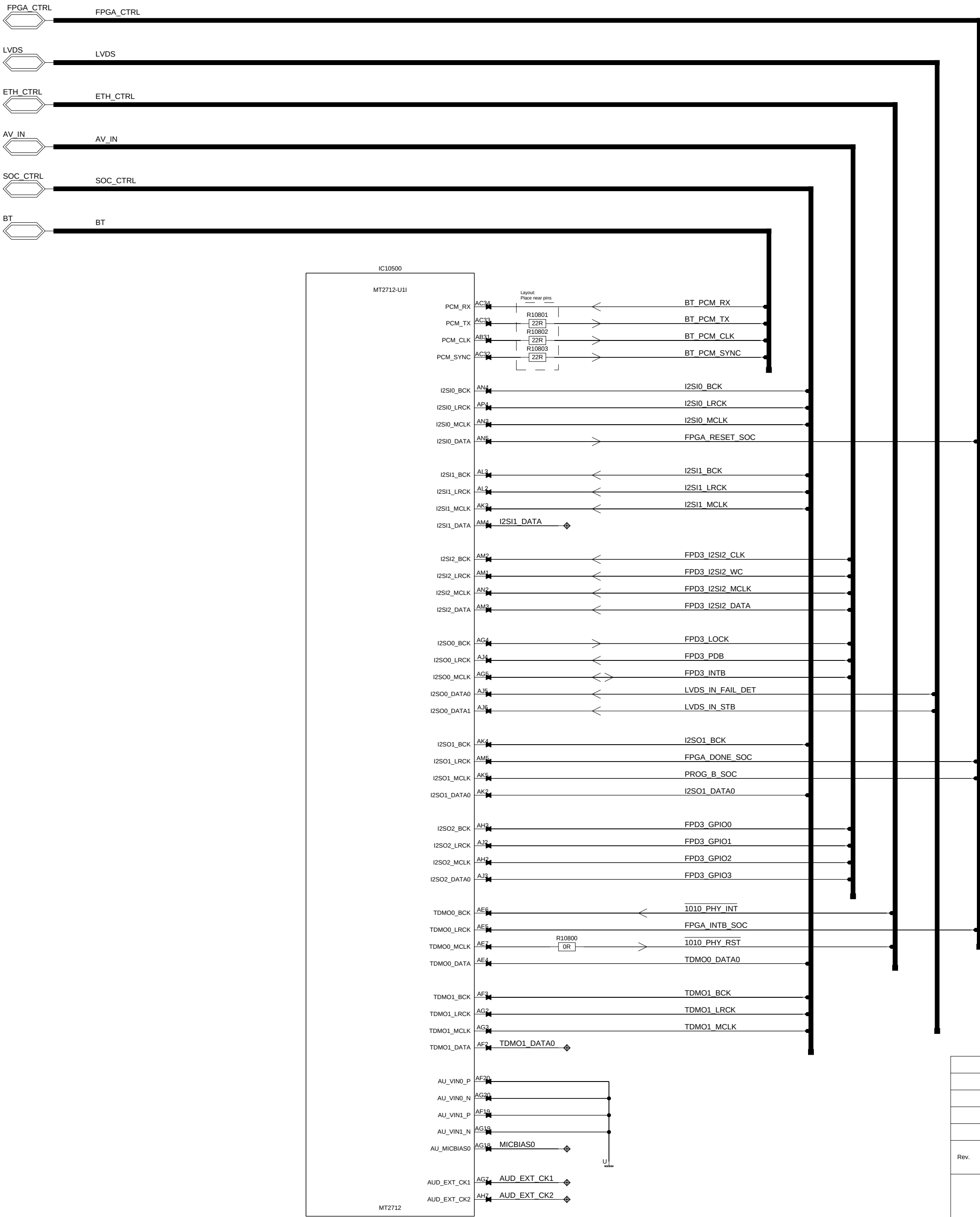


10200 - 10299

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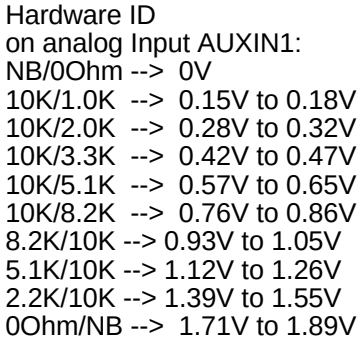
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4010_SoC			15	31		
substitute for:			Type	Variant		Part Number
substitute by:						

SOC_AUDIO (10800-10899)



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substitute for:				Type	Variant
				Part Number	

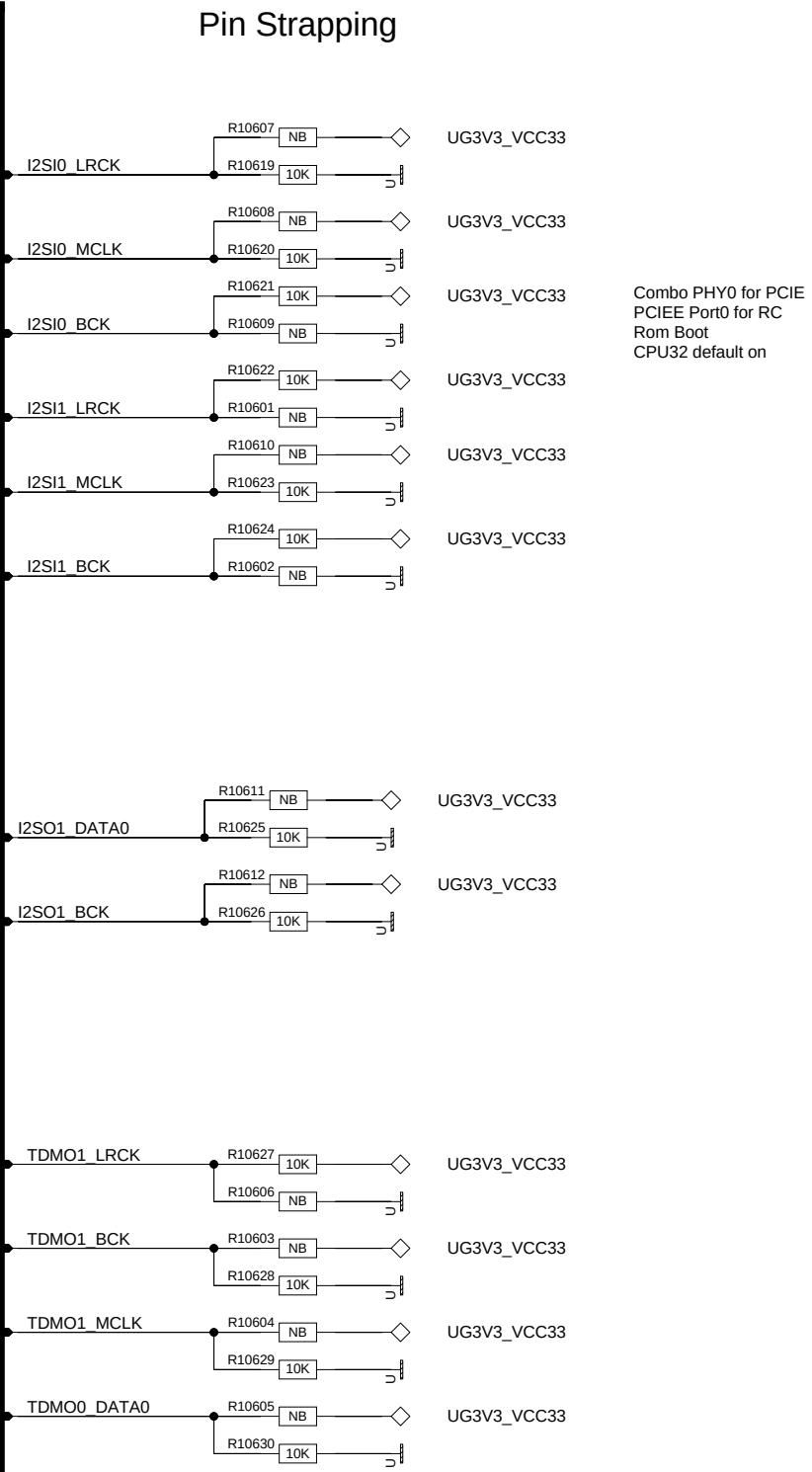
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A2 26.02.12

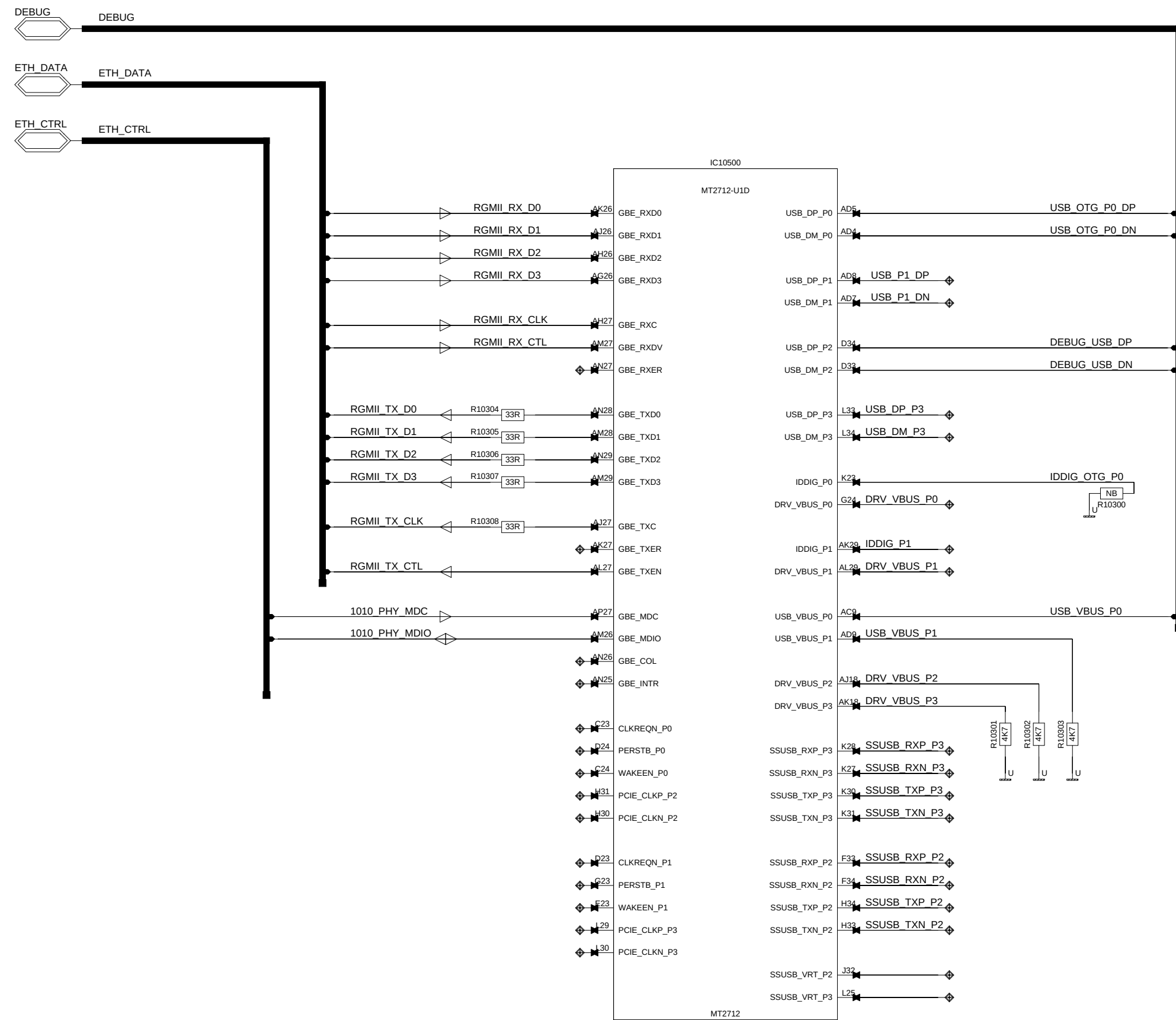
ISZO1_DATA0 status[1]	L	L	H	H
ISZO1_BCK status[0]	L	H	L	H
1	EMMC	NAND	NOR	SD
2	SD	EMMC	NAND	NOR
3	NOR	SD	EMMC	NAND
4	NAND	NOR	SD	EMMC

Trapping Mode	TDMM0_LRCK	TDMM0_BCK	TDMM0_MCLK	TDMM0_DATA0
Slowest Boot device speed configuration	1	X	X	X
Fastest Boot device speed configuration	0	X	X	X
NAND randomizer: disable	X	0	X	X
NAND randomizer: enable	X	1	X	X
NOR flash address mode: 3-byte	X	X	0	X
NOR flash address mode: 4-byte	X	X	1	X
Alternative boot loader: disable	X	X	X	0
Alternative boot loader: enable	X	X	X	1



Rev.	type of change	change-no.	date of change	auth.	checked
				date	name
		created			
		checked			
schematic		Sheet 17	Sheets of 31	Sample	
4010_SoC_CTRL		Type	Variant	Part Number	
substitute for:		substitute by:			

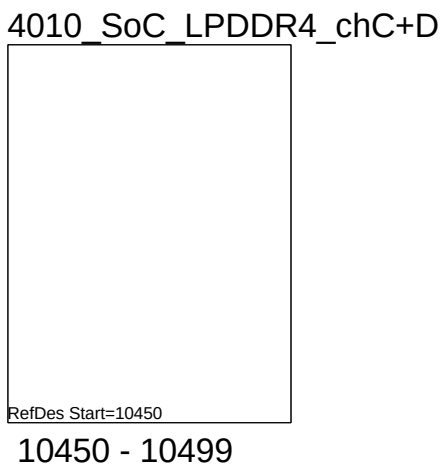
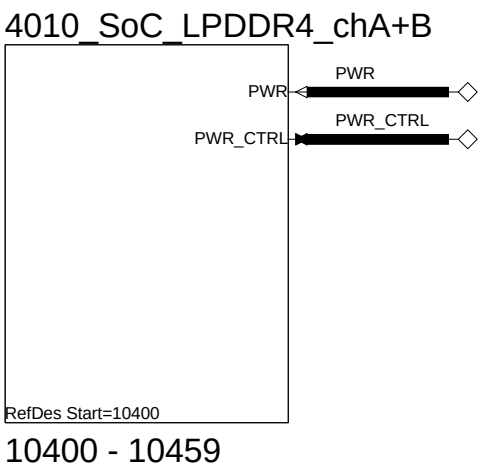
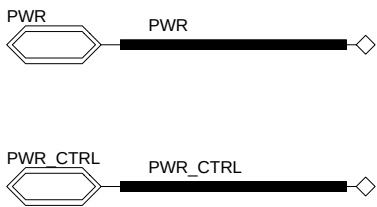
SOC_ETH_PCl_e_USB (10300-10399)



Rev.	type of change						change-no.	date of change		auth.	checked	
				Copyright reserved		Wiring modifications reserved						
								date		name		
								created				
								checked				
schematic								Sheet 18	Sheets of 31	Sample		
4010_Soc_ETH_PCIE_USB								Type	Variant	Part Number		
substitute for:				substitute by:								

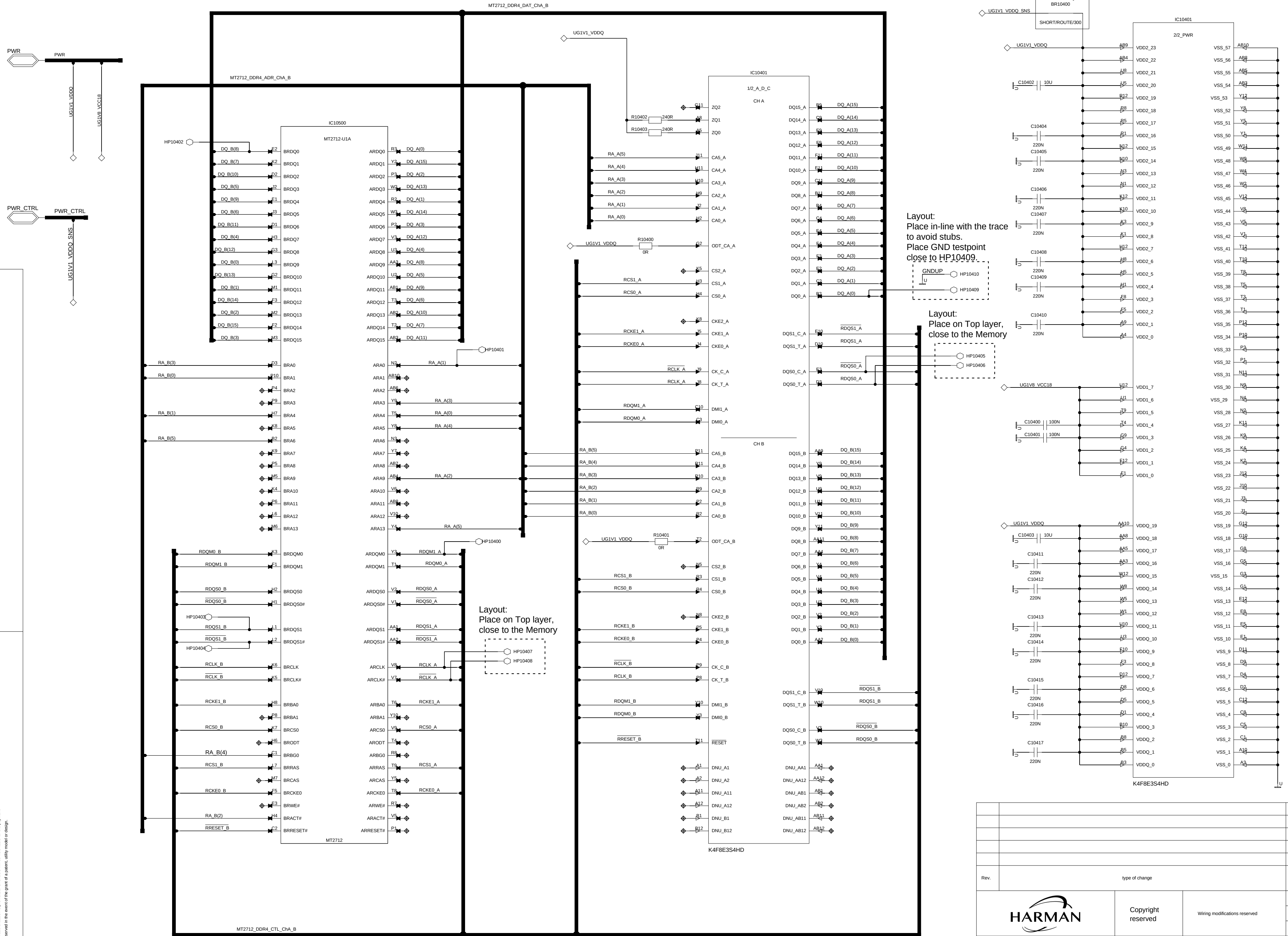
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SOC_LPDDR4 (10400-10499)



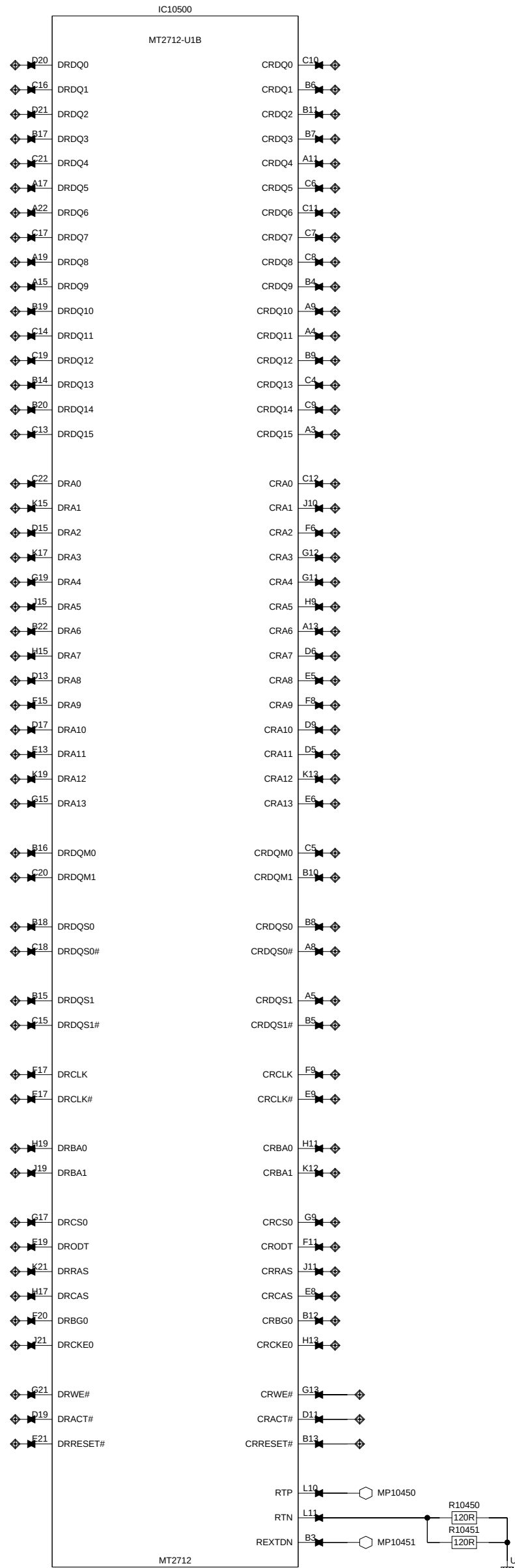
Rev.	type of change		change-no.	date of change	auth.	checked
		Copyright reserved	Wiring modifications reserved	date		name
				created		
				checked		
schematic			Sheet	Sheets of		Sample
4010_SoC_LPDDR4			19	31		
substitute for:			Type	Variant		Part Number
substitute by:						

SOC_LPDDR4 (10400-10449)



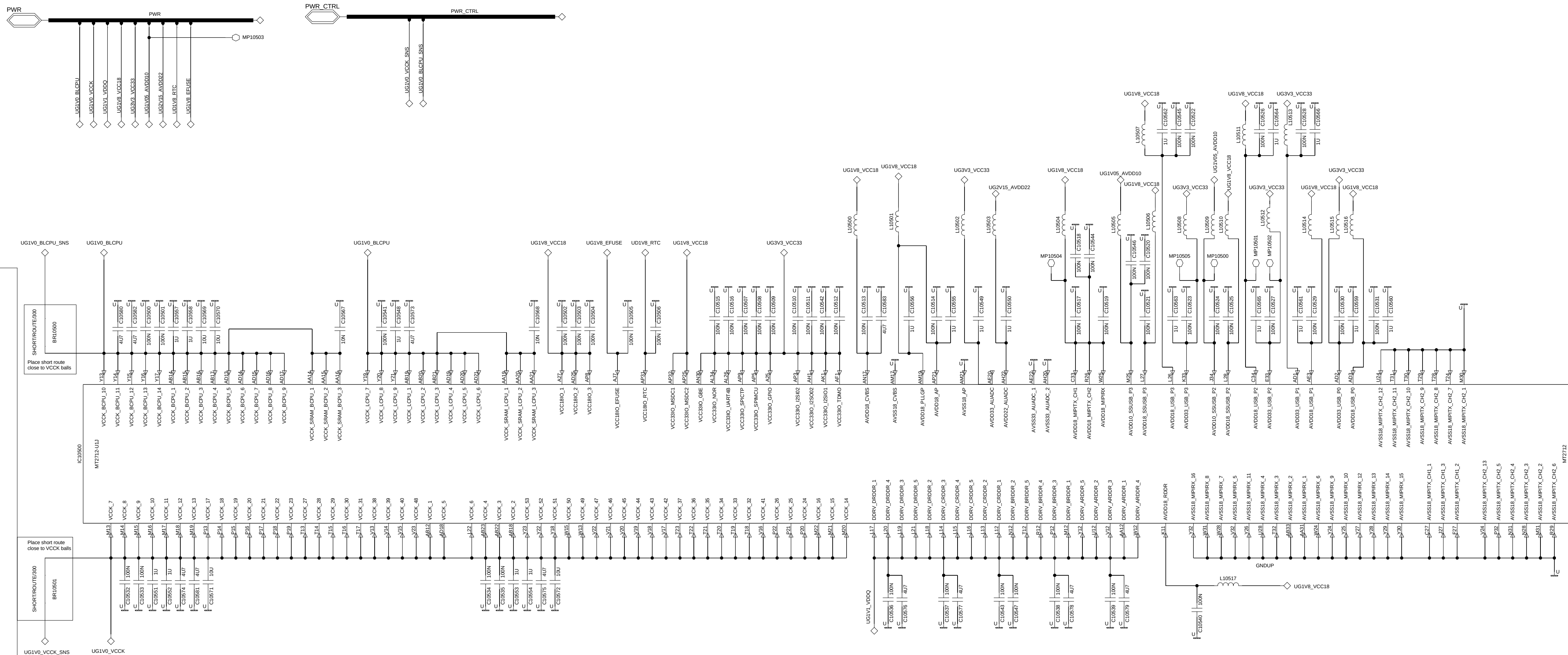
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SOC_LPDDR4 (10450-10499)

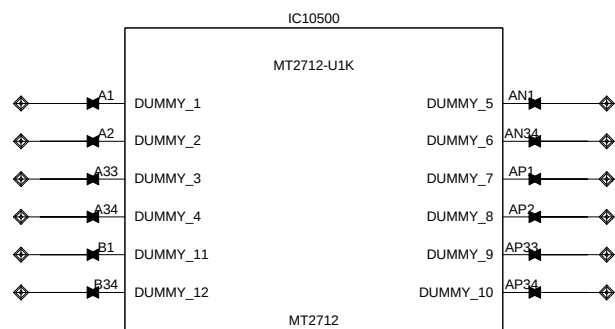


Rev.	type of change		change-no.	date of change	auth.	checked
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				created		
				checked		
schematic				Sheet	Sheets of	Sample
4010_SoC_LPDDR4_chC+D				21	31	
substitute for:		substitute by:		Type	Variant	Part Number

SOC_POWER (10500-10599)

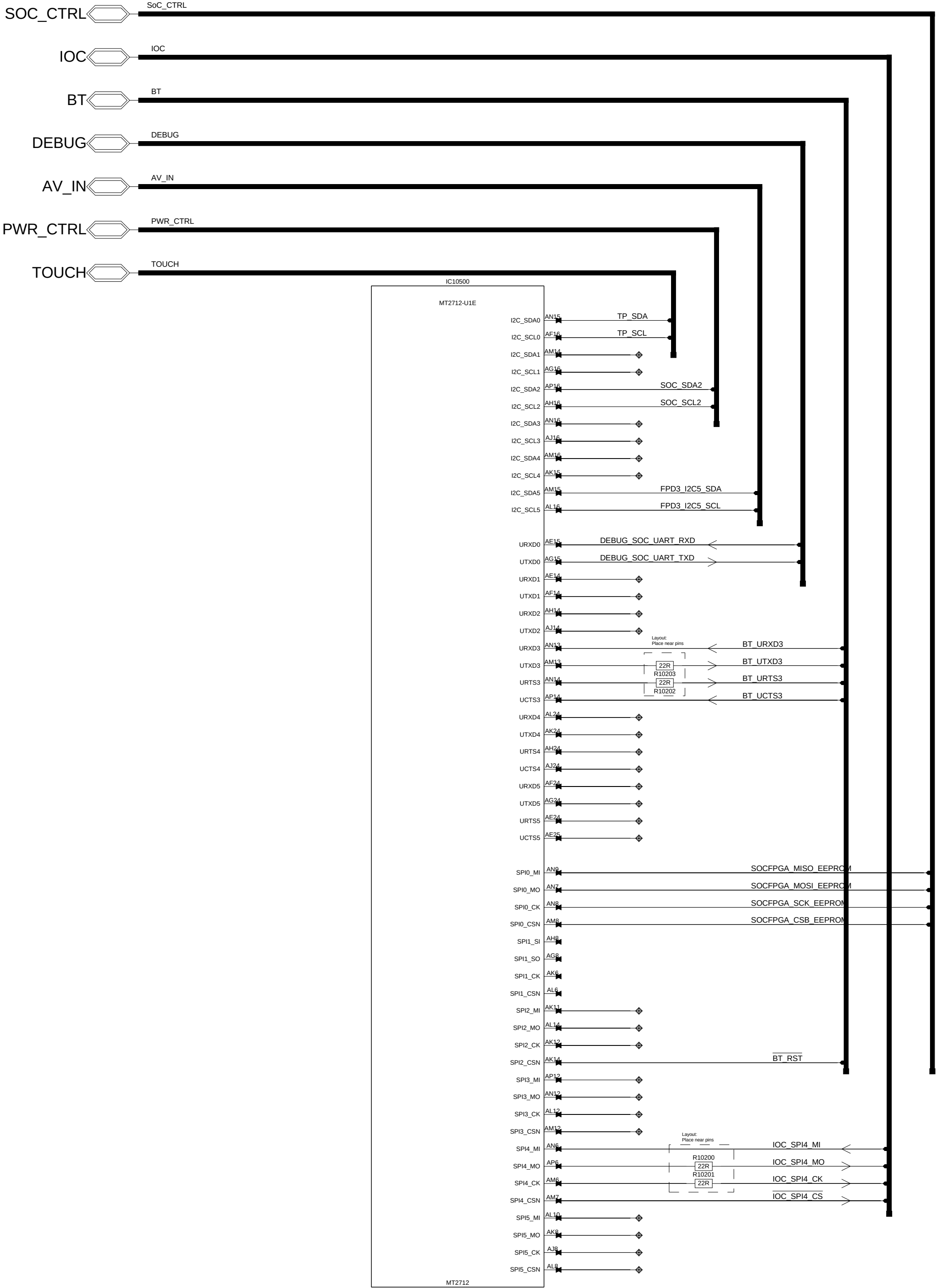


4010 SoC GND



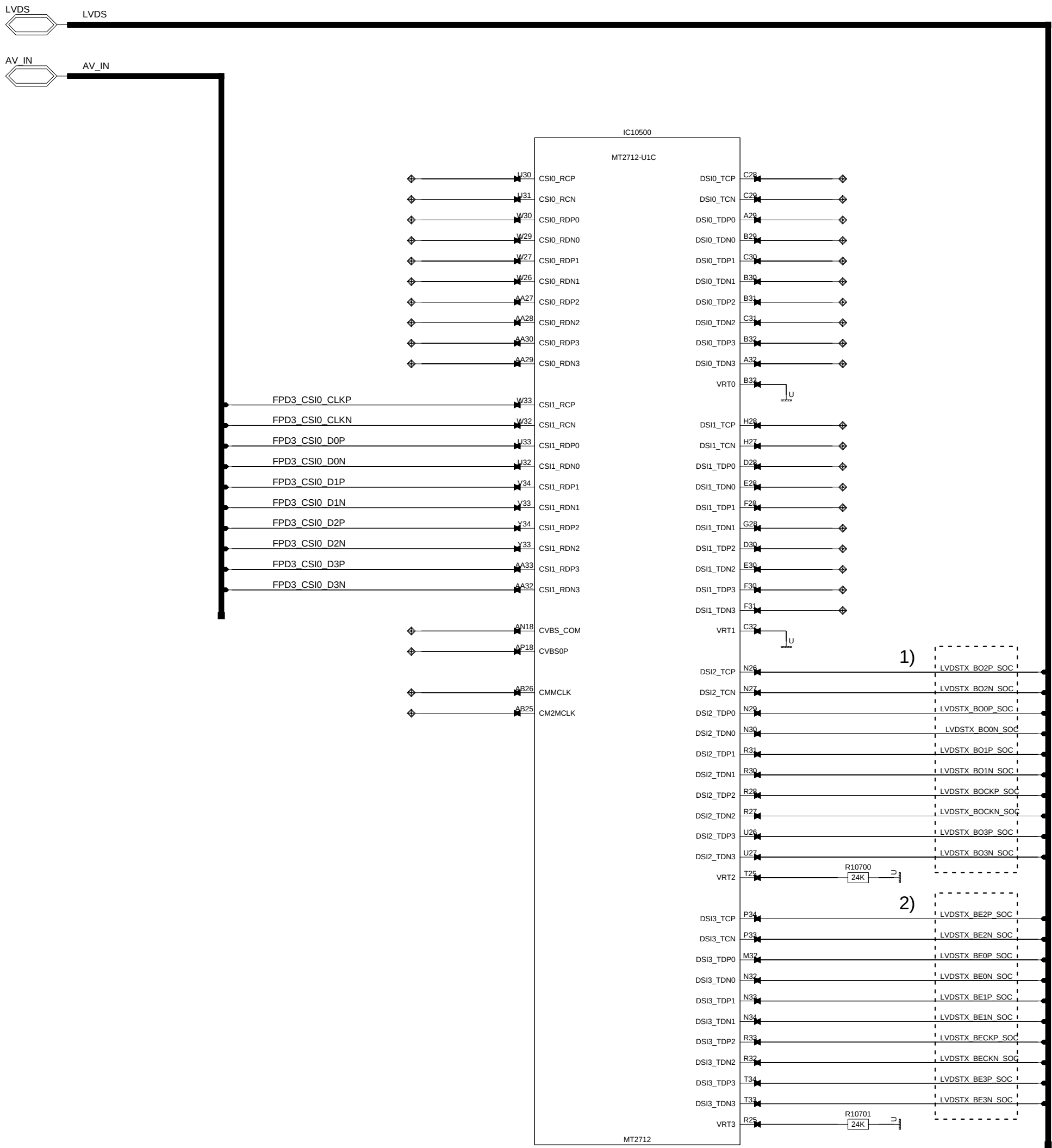
Rev.	type of change	change-no.	date of change	auth.	checked
	Copyright reserved	Wiring modifications reserved	date		name
			created		
			checked		
schematic		Sheet 23	Sheets of 31	Sample	
4010_SoC_GND		Type	Variant	Part Number	
substitute for:		substitute by:			

SOC_SERIAL_IF (10200-10299)



Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet 24	Sheets of 31	Sample
4010_SoC_Serial_IF			Type	Variant	Part Number
substitute for:		substitute by:			

SOC_VIDEO (10700-10799)



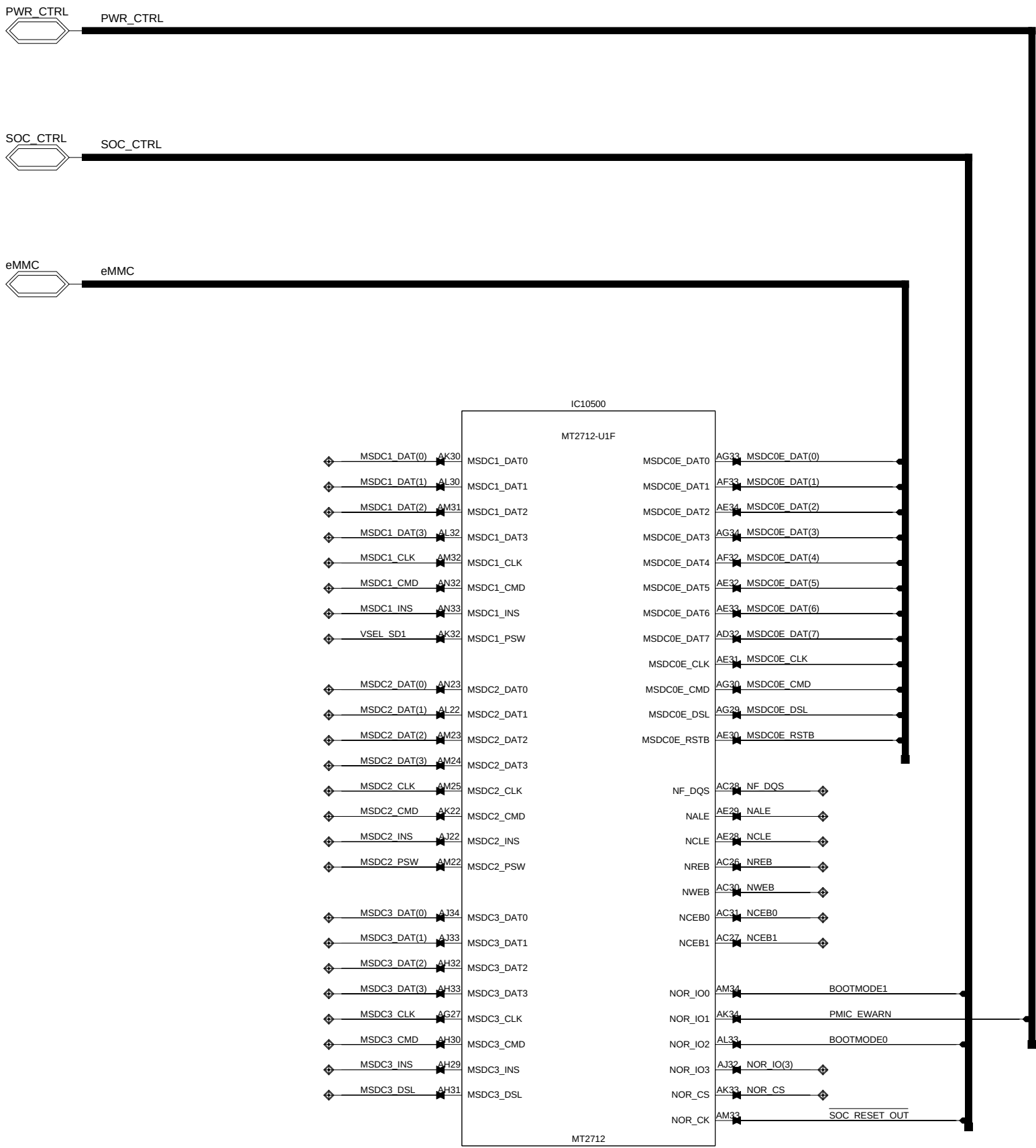
- 1) - Route as differential pair signals.
- Differential impedance: 100ohm (±10%)
- Maximum intra-pair skew: 0.127mm (5mils)
- Maximum skew from clock (BOCK) to data (BO[0-3]): 5mm

- 2) - Route as differential pair signals.
- Differential impedance: 100ohm (±10%)
- Maximum intra-pair skew: 0.127mm (5mils)
- Maximum skew from clock (BECK) to data (BE[0-3]): 5mm

Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet	Sheets of	Sample
4010_Soc_Video			25	31	
substitute for:			Type	Variant	Part Number
substitute by:					

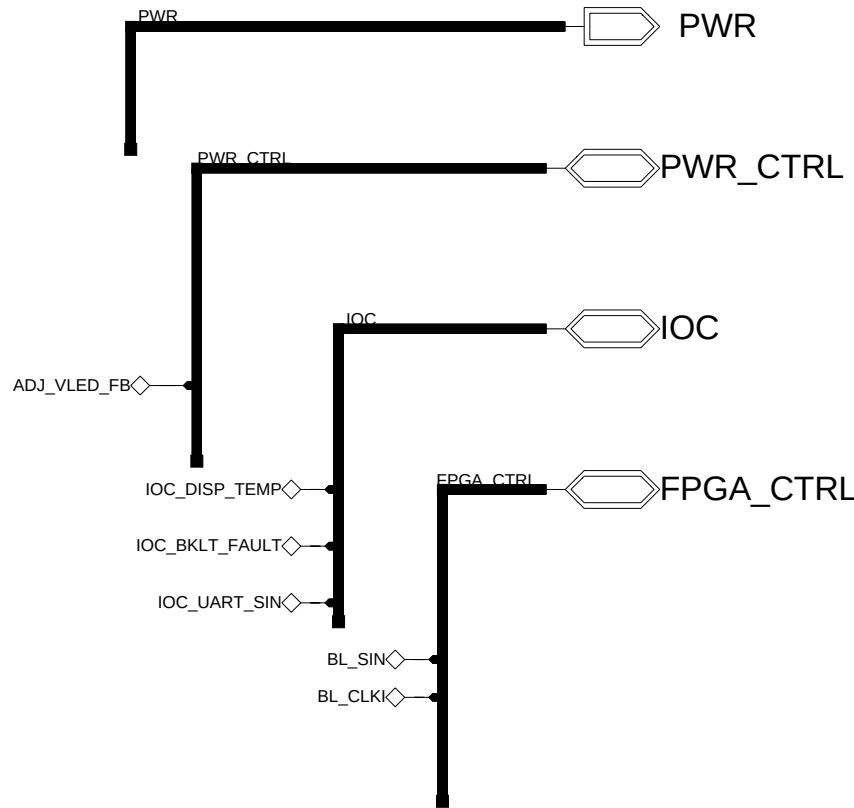
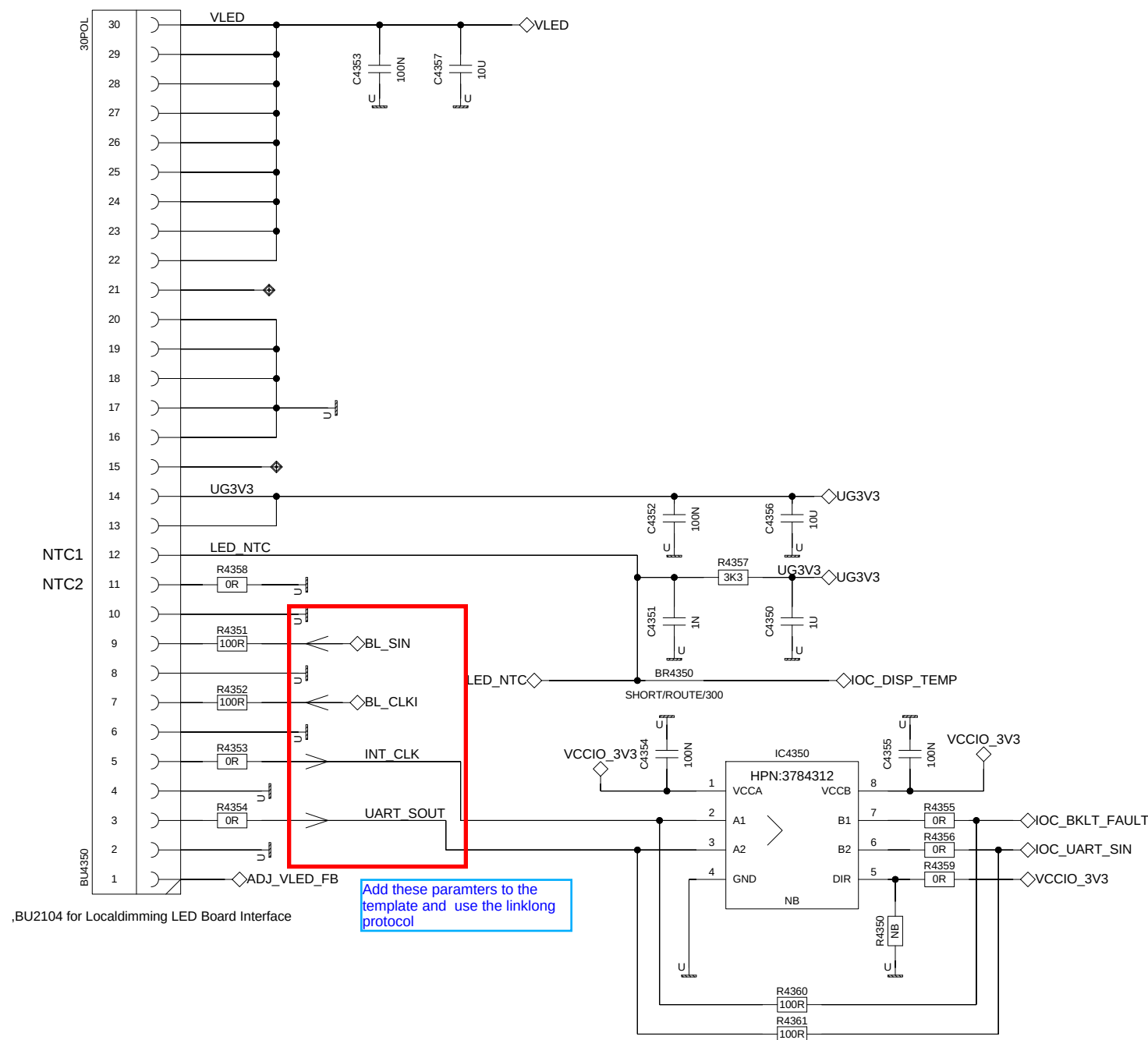
A2 00 02 12

SOC_eMMC_NAND_SID (10100-10199)

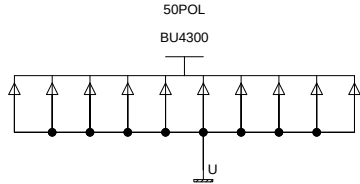
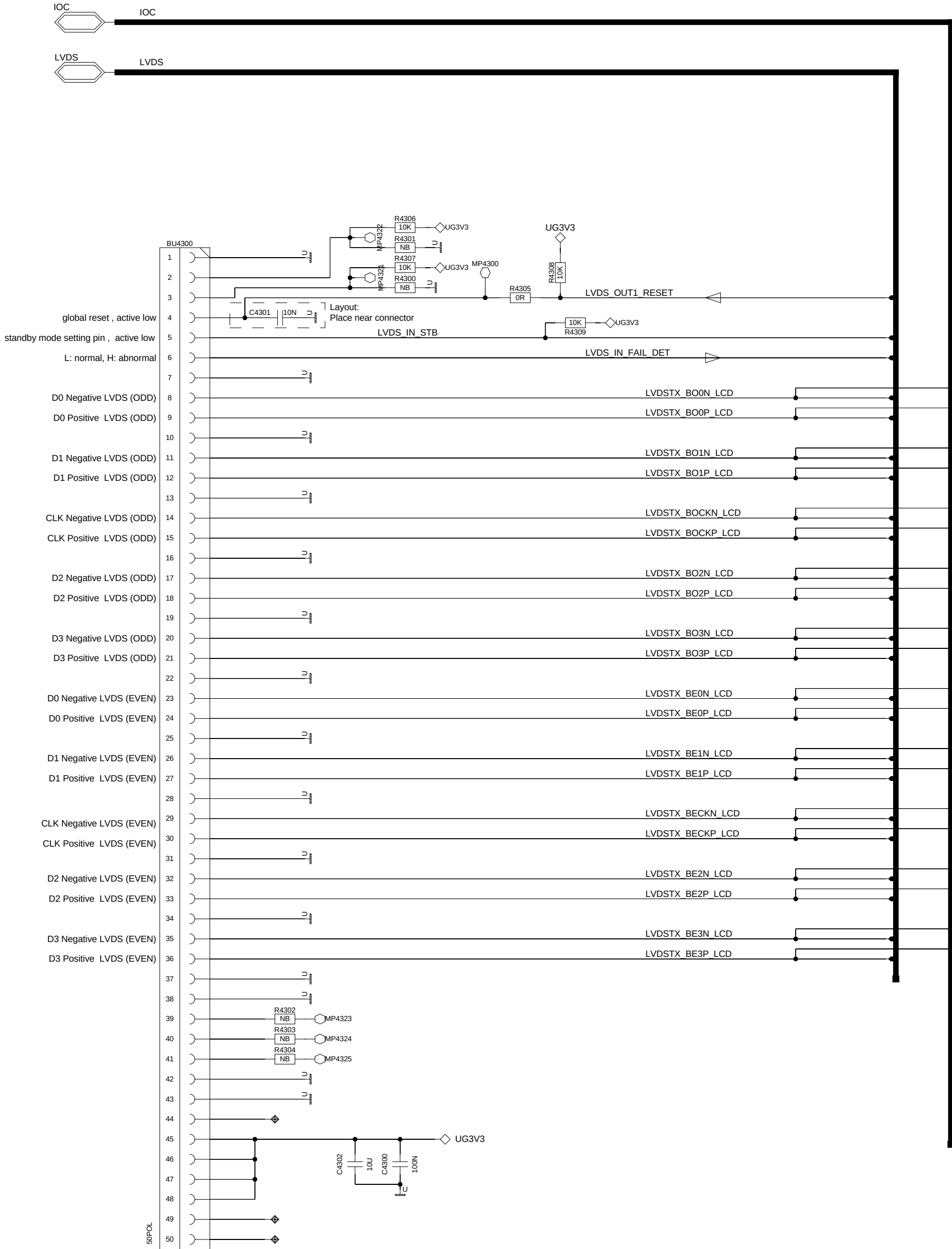
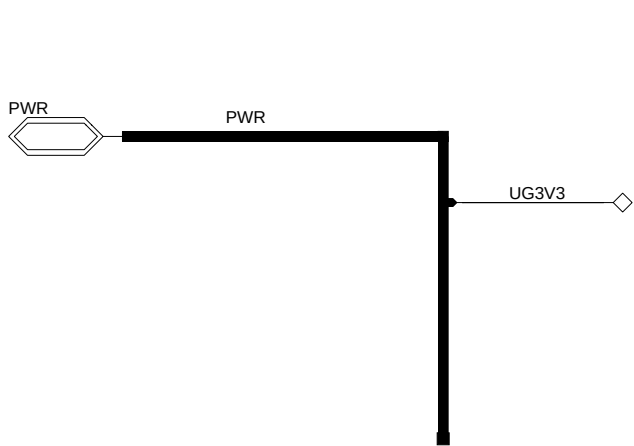


Rev.	type of change		change-no.	date of change	auth.	checked
		Copyright reserved	Wiring modifications reserved	date		name
				created		
				checked		
schematic				Sheet 26	Sheets of 31	Sample
4010_SoC_eMMC_NAND_SD				Type	Variant	Part Number
substitute for:		substitute by:				

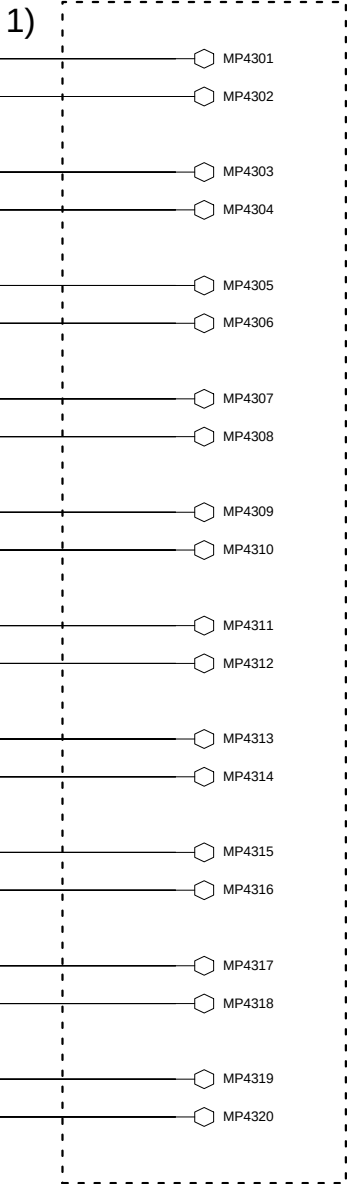
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	Copyright reserved	date		name	
		created			
		checked			
schematic	Sheet		Sheets of		Sample
4010_TFT_Bl_Conn	28		31		
substitute for:	Type		Variant		Part Number
	substitute by:		C143		



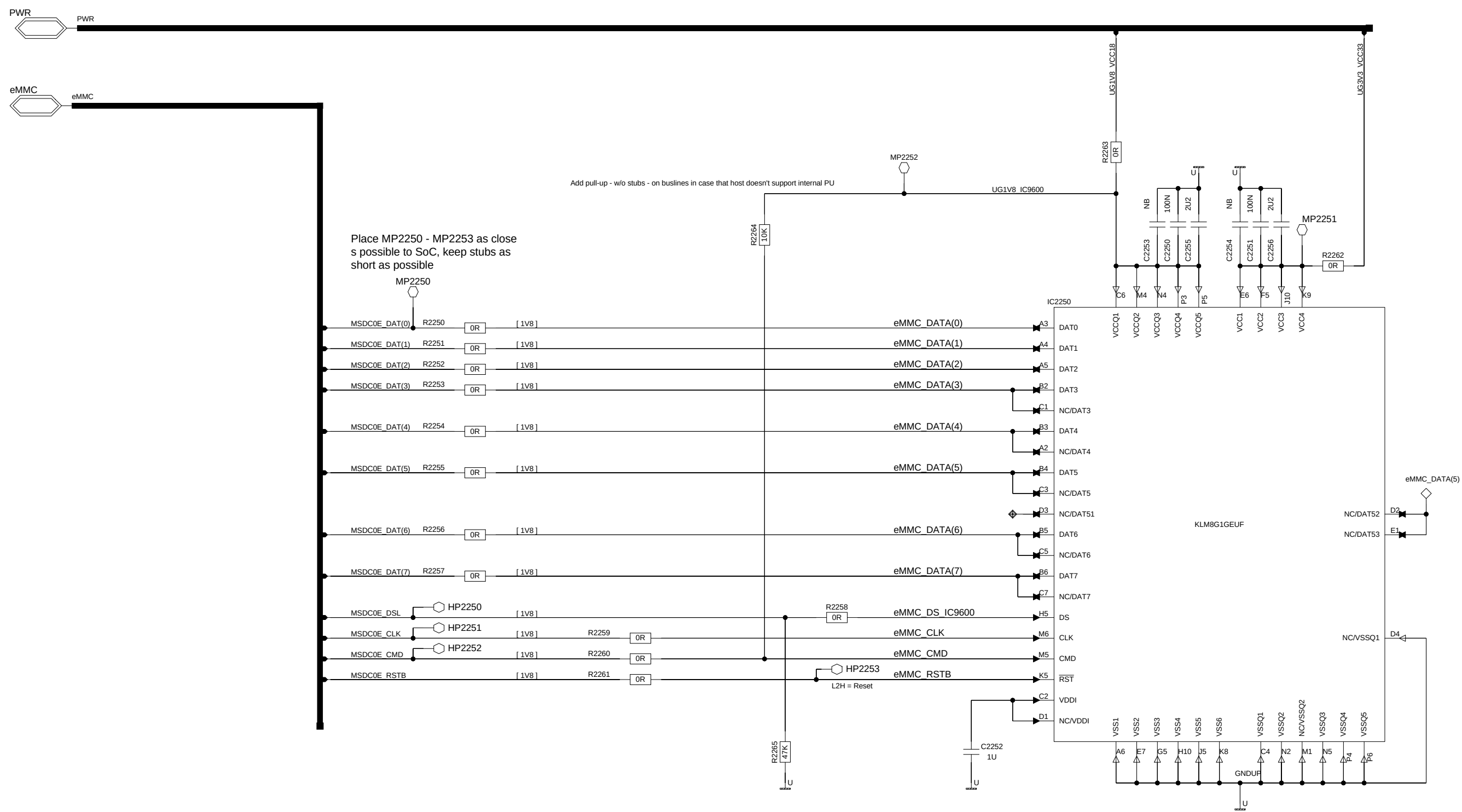
Verify pin out during layout



1) Testpoints should be placed as close as possible to an existing VIA

Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet 29	Sheets of 31	Sample
4010_TFT_Conn			Type	Variant C143	Part Number
substitute for:		substitute by:			

eMMC (2250-2299)



eMMC	Density	Part Number
Micron (-40 to 105°C)	8 GB	3881466
Micron (-40 to 85°C)	8 GB	3543617